Train on the Event, Deploy on the Baseline: The Change Interval Belongs to the Task Specification, not the Model

Luoyang Xiong, Juan Meng, Nan Deng, Wenchao Li, Xu Jin, Xiaotian Dong, and Yuchai Zhang

***Abstract*—Newly triggered landslides are mapped months after an event, because the annotations a supervised detector needs do not yet exist when a candidate set must be produced. Candidate discovery therefore carries two decisions that current practice leaves implicit: which route takes it, prompt-free proposals or a trained detector, and which two acquisitions are differenced when building the change channels. We settle both under one budget-equalized protocol that gives every method the same candidate budget, on four held-out events and ten cross-event and cross-region folds. Three results follow. First, the winning route is predictable from the data before any model is fitted: a criterion computed from the event's own masks reaches a Pearson correlation of 0.915 against measured recall, with the correct sign in ten of ten folds. Second, the change interval is a calendar fact rather than a tunable hyperparameter: bracketing the event date with each tile's own acquisition calendar reproduces the annotated window exactly in 97.9% of 3,135 tiles from 11 inventories, and within one acquisition in every tile. Third, the interval is committed during training and free at deployment: a frozen long-baseline model given the correct interval at inference falls from 0.7085 to 0.6217 AR@900, whereas committing the interval at training time while deploying the long baseline unchanged raises AR@900 from 0.7094 to 0.7474 over three seeds (+5.4% relative, +11.7% at 100 candidates), with no inference cost and no architecture change. The interval therefore belongs in the specification of the task, alongside the sensor and the resolution, rather than in the tuning of the model. The gain is sized on the event-level protocol; the calendar determination and the irreversibility of the interval hold under both protocols tested, and the size of the gain under a region-level protocol is reported separately.**

***Index Terms*—Landslide detection, region proposal, prompt-free detection, cross-event generalisation, multimodal remote sensing, Sentinel-1/2, transferability**

I. INTRODUCTION

A. Deployment is the condition that matters

Landslide inventories underpin disaster response, hazard assessment and risk planning, and they are needed within days of a triggering event, but they are usually assembled months later because annotation follows imagery; the Sentinel-1 and Sentinel-2 constellations now deliver global, free and day-and-night observations at a cadence far beyond what manual interpretation can cover. Newly triggered landslides, however, rarely carry immediate annotations, so the value of an automated mapping system rests on how well it transfers to a region, an event or a data source it has not seen [1]. The operational constraint is therefore not the quantity of imagery but the behaviour of the model under shift. That second question — at which stage of the pipeline the behaviour under shift is decided — is the subject of this paper.

B. Candidate discovery sets the ceiling

An automated inventory has a fixed order of operations: candidate discovery, verification, and mapping. The first stage is binding. A landslide that is never proposed cannot be recovered downstream, so the recall ceiling of discovery bounds the quality of the entire chain, and the first question about a discovery method is how many true landslides it places into a candidate set. The question this paper poses is whether that number can be anticipated rather than measured.

C. Two detector paradigms, and two dependencies that shift does not preserve

Two families of detectors dominate remote-sensing object detection. Closed-set detectors are trained to recognise landslides from labelled examples and internalise the appearance statistics of the events they saw. Open-vocabulary, text-conditioned detectors instead localise objects by matching image regions to class descriptions and are promoted as offering transfer without retraining. Distribution inconsistency between training and deployment settings is a long-recognised difficulty for large-scale landslide detection from multisource imagery [2], and recent work on cross-domain landslide mapping reports that even strong vision foundation models still generate high-confidence false alarms on unseen regions, events and data sources [1]. What the two families share is a dependence on something that shift does not preserve: event-specific appearance in the first case, and the reliability of a class name in the second. We compare the two dependences under a single candidate budget, equal for both.

D. The change interval as a second bottleneck

The change channels are the strongest single modality in our ablation, yet the convention that produces them is rarely

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Data and code availability: All result artefacts, split definitions, evaluation code and environment locks are provided in the accompanying repository/archive (_submission/, 1203-entry SHA256 manifest); an independently rebuilt copy of the dataset was verified SHA256-identical at the tile level, and its normalisation statistics recomputed bit-for-bit, before the reported runs. The underlying dataset (Sen12Landslides) is publicly available. The code, data definitions, per-image scores and result artefacts are openly available at https://github.com/xionglovetian/pfrpn-change-interval.

examined. The deployed setting differences the first and last acquisition of each tile, a span of 678 days on average, while the event itself occupies a single adjacent acquisition pair inside that span. The same convention is applied to unannotated tiles, because annotation-derived windows are not available for them. Three questions follow. Can the annotated window be recovered without any pixel-level annotation? Does a correctly positioned window carry more usable change signal? And if it does, where in the pipeline must it be applied?

The answers are ordered. The window can be recovered deterministically from the event date and the tile's own acquisition calendar. A correctly positioned window of one acquisition on each side separates landslide pixels better than the long baseline, while the single adjacent pair is worse than the long baseline because too few of its pixels are usable in both frames. Most importantly, the interval cannot be corrected at inference time: a frozen model trained on the long baseline degrades when it is given the correct interval. The correction must be committed during training, and once it is, the deployed input can stay exactly as it is. Training interval and deployment interval are therefore two independent choices that the literature treats as one. This paper answers both: the interval is recoverable without annotation, and the pipeline stage where it must be applied is fixed by training rather than by inference.

E. The comparison this literature has not made at the proposal stage

Transferability research in landslide remote sensing has largely addressed segmentation models and cross-region settings [3]; we are not aware of a study that compares detection paradigms at the proposal stage under event- and region-disjoint protocols, which is the gap addressed here. Independently, PF-RPN proposed a prompt-free universal region proposal network in early 2026 [4], replacing exemplars, predefined categories and text with a purely visual, class-agnostic proposal mechanism. Because such a network need not identify what an object is in order to localise it, it is a natural candidate for the discovery stage of landslide mapping. To our knowledge PF-RPN has not been applied to landslide imagery which leaves its behaviour under event and region shift as the open question this paper takes up. Prompt-free and parameter-efficient adaptation has already been explored in this domain, but at the segmentation level: variants of the Segment Anything Model have been adapted to landslide segmentation with multisource inputs and small target-domain label counts [5], [6], [7], [8], and cross-regional few-shot segmentation has been benchmarked on a large multi-region dataset [9]. None of these studies addresses the candidate-discovery stage, compares paradigms under a reported per-arm candidate budget, or asks whether the outcome of such a comparison can be anticipated before any model is trained. Whether the paradigm survives where the two dominant detector families lose most of their recall, and whether that can be known in advance, is therefore an open empirical question rather than a settled one.

F. This study and its contributions

We answer four questions in order: whether the winning route can be forecast from the data without training, whether the annotated interval can be recovered without annotation, whether a correctly placed window carries more usable change signal, and where in the pipeline that window must be applied. The contributions of this study are five.

(1) The change interval is a training specification, not an inference setting. Committing the interval during training while leaving the deployed long baseline unchanged raises AR@900 from 0.7094 to 0.7474 over three seeds (+0.0380, 5.4% relative, 5.5 standard errors), growing to +11.7% at 100 candidates per image, at no inference cost and with no architectural change. The order matters: giving the correct interval to a frozen long-baseline model instead lowers AR@900 by 0.087 (0.7085 to 0.6217), and the interval itself is available from the calendar before training begins (contribution 2). The gain is measured under the event-level protocol; a leave-one-region-out fold does not reproduce it, which is the scope of this claim rather than of the specification (Section VI-E).

(2) The interval as a calendar fact. Bracketing the event date with the tile's own acquisition calendar reproduces the annotated pre/post window exactly in 97.9% of 3,135 tiles across 11 inventories, and within one acquisition in 100.0%. The interval is therefore available at deployment with no annotation.

(3) A training-free criterion that selects the route before training. A cheap property of the imagery, computable wherever reference masks exist, predicts which candidate-discovery route will win on an unseen event, with a Pearson correlation of 0.915 against measured recall, a leave-one-fold-out correlation of 0.856, and the correct sign in ten of ten folds.

(4) A budget-equalized cross-paradigm comparison at the candidate-discovery stage. Under event- and region-isolated protocols, with every arm's real candidate budget reported, the paradigm gap is consistent and quantified per arm (2.13x on the four-event split, 3.11x over ten folds; both are image-averaged AR@900 at a matched 900-candidate budget), and part of the magnitude reported for this task reflects the candidate budget of the arms rather than the paradigm.

(5) A single-variable intervention suite and a score-semantics-neutral verifier. The suite separates information removal from distribution shift, and interval position from temporal-baseline span. The verifier makes arms with incomparable native scores rankable, which is what makes a cross-paradigm contrast measurable at all.

II. RELATED WORK

A. Cross-domain landslide mapping with vision foundation models

Recent work has reframed cross-domain landslide mapping as a problem of adapting a frozen visual backbone rather than of building a new one. GeoPhysAdapter anchors on a frozen vision foundation model and restricts terrain, material and triggering information to dense guidance, regional modulation and event-timing forcing, applying bounded adaptation at the pixel and candidate-body decision units and reverting to the

visual prediction where geophysical support is insufficient; on an event-isolated multi-source benchmark it reports an error reduction of 7.76% at the pixel level and 23.99% at the candidate-body level [1]. That study establishes two premises we adopt. First, cross-domain transferability, not in-domain accuracy, determines operational value, because newly triggered landslides lack immediate annotations. Second, an evaluation is only interpretable if held-out units are isolated, which motivates the event- and region-disjoint protocols used here. Our aim differs in one respect: rather than proposing an adapter, we ask whether the *choice of detection paradigm* determines whether the discovery stage survives shift at all.

The dominant line of work detects landslides directly with fully supervised models: Mask R-CNN applied to high-resolution optical imagery [11], densely connected networks combined with environmental conditioning factors [12], and controlled comparisons of CNN, ResNet and DenseNet backbones [13]. Later work has traded accuracy for efficiency [14], [15], injected prior knowledge such as susceptibility [16] and shape [17], or reformulated the learning objective itself [18], alongside applications to single triggering events [19]. Two features recur. First, evaluation is normally conducted on one or two study areas with a within-event split, so reported accuracies describe performance inside an event rather than transfer to the next one [11]–[13]. Second, the object of study is the model class, which makes each conclusion a statement about its own architecture.

B. Cross-event evaluation in landslide modelling

Leave-one-event-out validation is established practice in landslide modelling. Global earthquake-triggered landslide prediction was assessed over several decades of strong earthquakes with leave-one-event-out cross-validation and an independent cross-continental test spanning diverse climates and faulting styles, and that study reported that incorporating non-triggering earthquakes markedly improved cross-event performance [10]. The protocol we use is therefore not novel in kind. Less developed is its use to compare detection *paradigms* at the proposal stage, where the quantity of interest is a recall ceiling rather than a segmentation or susceptibility score.

Systematic evaluation of what does *not* survive a shift is itself an established genre in this field: controlled studies of domain generalisation [20], [21], benchmarking protocols for an entire task family [22], and experimental studies whose conclusion is that a simpler default is no worse than the elaborate alternatives [23]–[25].

C. Text-conditioned, open-vocabulary and prompt-free detection

Open-vocabulary detectors localise objects by aligning image regions with textual descriptions [26], [27] and have been applied to remote sensing with claims of transferability to unseen categories. Their limitation under domain shift is the subject of an emerging critical literature: in a different domain, a systematic evaluation of two dominant detection paradigms (training-free and supervised) reported that both were brittle under distribution shift [28]. The structural lesson we take from that study is the shape of the argument rather than its subject matter: an evaluation design, a demonstration that both candidate paradigms fail, an attribution to the paradigm rather than to the task, and an explicit statement of where the conclusion stops. Cross-domain few-shot object detection in remote sensing has meanwhile framed the deployment constraint as a domain gap and addressed it with architectural adaptation [29]; that framing is useful for the problem statement, though the contribution there is a method rather than a paradigm-level diagnosis. Landslide4Sense, finally, has provided a large multisource benchmark and framed distribution inconsistency as a central difficulty for large-scale landslide detection [2]. The proposal lineage itself descends from class-agnostic region proposal networks [30], which localise candidate objects before any semantic decision. Class-agnostic proposals have already been shown to transfer better than category-conditioned ones: a classification-free object localisation network localises novel objects absent from its training categories, precisely because the binary classifiers of existing proposal methods overfit to the training categories [31]. Our setting differs in the shift that has to be crossed — not a set of unseen categories, but an unseen event, in which lithology, vegetation and triggering mechanism all change at once. Within remote sensing, open-vocabulary detection and segmentation have advanced through distillation from vision-language models [32], adapters that improve their generalisation [33], and foundation models pretrained on multiple tasks [34]. The limits of the family are nevertheless documented: a systematic evaluation of a recent general model reports robust behaviour on visible imagery but marked degradation in infrared and SAR [35], and cross-domain few-shot detection remains limited when inter-domain discrepancy is left unmodelled [36]. Our study sits between these lines: we adopt a prompt-free, class-agnostic proposal paradigm, measure its recall ceiling under strict event and region isolation, and compare it against both detector families rather than against a single method.

Within landslide remote sensing the same adaptation move has appeared at the segmentation level. Prompt-free and parameter-efficient variants of the Segment Anything Model have been applied to landslide imagery [5], [6], [8], with terrain-aware fusion across levels [6] and frozen encoders carrying lightweight adaptation modules [8]; LSDSAM couples a task-specific branch with three-step transfer learning over Landslide71K to reach reliable detection from single-digit labelled samples in a target domain [7]; and cross-regional few-shot segmentation has been benchmarked on a large multi-region dataset with a frozen visual backbone [9]. These studies report accuracy gains from adaptation on one or a few target areas. None evaluates the candidate-discovery stage, none compares paradigms under a reported per-arm candidate budget, and none asks whether the comparison can be forecast without training. The evaluation design below is built for those three questions.

III. DATA AND STUDY PROTOCOL

A. Imagery and tiles

We used Sen12Landslides [37], a publicly released multi-source, multi-temporal benchmark covering eleven landslide

events across diverse lithological and climatic settings. Each tile combines Sentinel-2 optical imagery, Sentinel-1 SAR, a Copernicus DEM, and pre- and post-event acquisitions that permit an explicit temporal-change representation. Tiles were processed at their native 128 × 128 pixel footprint and carry both bounding-box and pixel-level annotations. Every arm is trained and evaluated at 320 × 320 pixels, obtained by upsampling the native grid identically for all modalities; this keeps the comparison fair but adds no spatial detail, and the smallest annotated objects therefore remain only a few pixels across. The overall study design is summarised in Fig. 1.

Landslide-bearing tiles were taken from the S12LS-LD task (landslide-bearing tiles) and hard negatives from S12LS-AD (hard-negative tiles), the two subsets released with the benchmark, so that the model is exposed to background visually confusable with landslides rather than to easy negatives only.

Other public landslide benchmarks exist and differ in coverage — a multisensor collection of 20,865 images spanning nine regions [38], and inventories assembled for single triggering events [19]. Sen12Landslides is used here because it pairs Sentinel-1 and Sentinel-2 acquisitions with event labels, which the event- and region-isolated protocols require.

B. Modalities and the change channel

Four input modalities were used: Sentinel-2 pre-event optical reflectance; the ascending Sentinel-1 stack, chosen for consistency across events; DEM-derived topography; and an event-change representation. The event-change modality was defined on a long baseline — for every tile, the first and last acquisitions were differenced to yield three channels (dNDVI, dNDWI, dNBR) irrespective of the annotation. This definition is deliberate. An annotation-derived event window cannot be applied to negative tiles, which carry no annotation, and using different rules for positive and negative tiles creates a shortcut between the change channels and the label. The long baseline is label-free and therefore identical for positive and negative tiles. Normalisation statistics were computed on the training split only.

Every change channel in this paper differences a pre-event and a post-event acquisition. The deployed setting, which we call the long baseline, takes the first and last acquisition of each tile, a mean span of 678 days. The annotated window is instead a single adjacent acquisition pair, with a median span of one pair, located near the middle of the series. We reproduce that window independently from the event date and the tile's own acquisition calendar, choosing the last acquisition before the event and the first after it. Over 3,135 annotated tiles from 11 inventories the window is recovered exactly in 97.9% of tiles and within one acquisition in 100.0%. The median distance from the event to the bracketing acquisitions is 3 days for the previous one and 4 days for the next, with a 90th percentile of 7 days. The event interval is therefore determined without any pixel-level annotation, and a neighbouring event's calendar can stand in for unannotated tiles. For multi-event tiles, whose dating module returns a confidence-ranked set of candidate dates, we use the highest-confidence candidate. Recomputed long-baseline stacks match the deployed arrays exactly: over 40 tiles from the four test events the maximum absolute difference is 0. (see S1-F2 and S1-F3).

C. Protocols

Three partitions were used, and no event appears in more than one subset of a given protocol, so every reported test event was absent from training. The event-level split assigns five events to training (italy, kyrgyzstan1, hiroshima, kyrgyzstan2, itogon), two to validation (newzealand, hokkaido) and four to testing (chimanimani, indonesia, china, usa). Leave-one-event-out (LOEO) uses seven folds, each holding out one event entirely; the four remaining events (china, usa, itogon and kyrgyzstan2, with 72 to 220 tiles each) are used for training only, and the seven folds span 284 to 4,236 tiles. Leave-one-region-out (LORO) uses three folds (East Asia, Central Asia, South-East Asia), each holding out an entire region that is absent from training. Because no event straddles subsets, LORO is the protocol closest to deployment. Fold definitions are released with the manuscript. The paradigm contrast was evaluated under all three protocols: all five external baselines on the event-level split, and the three paradigms (closed-set, text-conditioned and prompt-free) across all ten cross-setting folds under a common cap of 900 candidates per image, with the realised candidate count of every arm reported. The proposal route and Grounding DINO fill that cap on every image, but the fine-tuned detector reaches a median of 359 to 900 candidates per image (it fills the cap on one of the ten folds) and the closed-set arm a median of only 118 to 296, so the fold-level comparison between paradigms is read at the strict common budget of 100 candidates per image, where every arm is matched; AR@900 is reported alongside as the ceiling each arm actually reaches.

D. Metric and reporting rationale

Candidate discovery is evaluated by average recall at k proposals per image (AR@k), k = 100, 300 and 900, at IoU 0.5. AR@k is the mean of per-image recall over the images that contain ground truth, so images without ground truth are excluded rather than counted as zero; it is not ground-truth weighted, and the ground-truth-weighted counterpart is reported alongside it as a micro value wherever arms are compared, because the two differ by 0.03–0.05 AR on identical proposals (0.3321 against 0.2900 for the fine-tuned YOLO-World at 900 candidates, a difference of 0.042). Table S1-C1 reports both conventions for every arm at all three budgets. Unless a micro value is named explicitly, every paradigm comparison in Section V is image-averaged. AR@k is a recall-ceiling measure: it reports how many true landslides fall inside a fixed proposal budget, which is the quantity the downstream chain inherits. FPPI-based operating-point metrics were not used because the adapted network assigns proposal scores in a very narrow range (approximately 0.31-0.51), so recall at low false-positive rates is uninformative at this stage; this is a property of the proposal stage and is reported so that the metric choice is transparent.

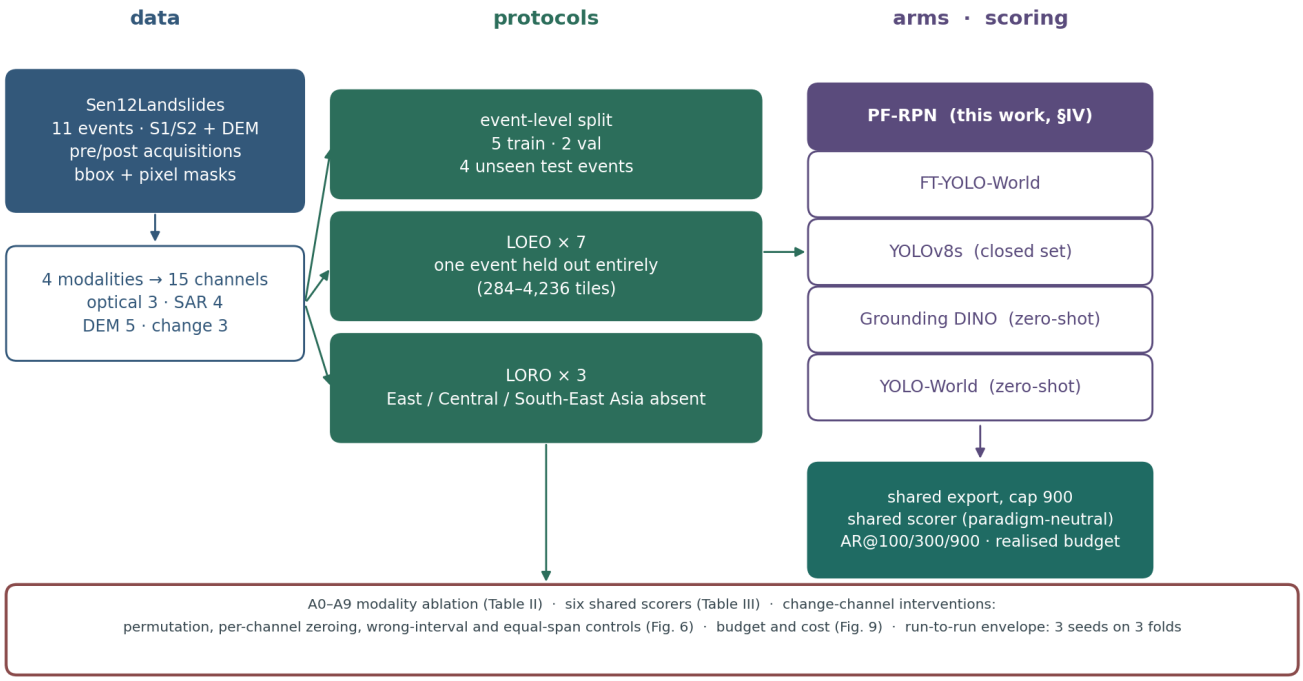


Fig. 1. Study design. The eleven-event benchmark supplies four modalities (15 channels) and the pixel masks; three event- and region-disjoint protocols are used; five arms are scored through one shared export and one paradigm-neutral scorer, with the realised candidate budget reported for every arm; the analyses listed at the bottom of the figure are the ones reported in Sections V and VI.

IV. METHOD: ADAPTING PF-RPN TO LANDSLIDE IMAGERY

A. PF-RPN as the adopted paradigm

The adopted base is a two-stage, DINO-style grounding detector [4]: a Swin-B backbone, a six-layer encoder and a six-layer decoder carrying 900 object queries, dynamic denoising during training, and a head that predicts a class-agnostic objectness, a box, a centre-ness and an IoU for every query. Two changes make it prompt-free. The text branch is removed, so no class name is an input, and the learnable text embedding is replaced by three modality-aware pseudo-tokens predicted from the imagery, so that the query conditioning comes from the scene rather than from a word. The number three is not a free hyperparameter: the learnable text embedding of the released checkpoint has exactly three rows, and a sparse mixture-of-experts router over the four pyramid levels selects the three best-scoring level features and mixes them into those three embeddings. Because the head is class-agnostic and its class score is never used for ranking in any experiment reported here, the network is used exactly as a candidate generator, which is the role assigned to it above. All inherited parameters are initialised from the released PF-RPN checkpoint (Swin-B, pre-trained on COCO and ImageNet); the modality branches, the fusion module and the pseudo-tokens are new and are trained from scratch on this benchmark.

B. Four-modality input

A tile is a 15-channel tensor: pre-event optical reflectance (3 channels), the ascending Sentinel-1 stack (4: VV and VH, pre- and post-event), topography (5: DEM elevation, slope, aspect, curvature and the topographic wetness index) and the event-change representation (3: dNDVI, dNDWI, dNBR). Per-channel means and standard deviations are computed on the training split only and applied as a single 15-channel normalisation, and the native 128 $\times$ 128 tile is resampled to 320 $\times$ 320 identically for every modality. Each modality is encoded separately before any mixing, and the branches are deliberately asymmetric: optical uses the Swin-B of the base (depths 2-2-18-2, heads 4-8-16-32, window 12) initialised from the released checkpoint; SAR uses a separate Swin-T (depths 2-2-6-2, heads 3-6-12-24, window 7) trained from scratch; topography and change each use a lightweight convolutional stem that maps its channels to the same 256/512/1024 pyramid. Fusion then happens at every pyramid level. A 1 $\times$ 1 convolution projects each branch to the level width, a level-specific gate of the form Conv-GELU-Conv emits one weight per modality, the weights are softmaxed over the modalities present in the sample, and the fused level is the weighted sum of the projected features. Two details matter for what follows. The gate heads are zero-initialised, so training starts from the optical-only behaviour and the gate learns how much of each modality to admit; and a per-sample modality mask drives the gate of an absent modality to zero, which is what allows the SAR- and change-ablated variants of Table II to be switches on one architecture rather than separate networks. Each fused level also emits a pooled modality token. These four tokens, together with the three pseudo-tokens, are the conditioning signal that replaces text (Fig. 2).

C. Training

The objective is a weighted sum of four head losses and three regularisers, given in displayed form below (plus a router load-balancing term and its z-loss, 0.01 and 1e-3, in the configurations that enable the router).

$$L = L_{cls} + 5L_{box} + 5L_{ctr} + 2L_{giou} + 0.01L_{align} + 0.01L_{miss} + 10^{-3}L_{div} \quad (1)$$

The head terms are a focal objectness loss (alpha = 0.25, gamma = 2.0), an L1 box loss, an L1 centre-ness loss and a GIoU loss on the final boxes; L_align keeps the per-modality tokens of the same tile close to one another; L_miss penalises drift in both the fused features and the modality tokens when a modality is dropped, and is trained with modality dropout p = 0.1 so that the representation does not rely on every modality being present; L_div keeps the three pseudo-tokens from collapsing onto one direction. The full network has 211.4 M parameters (86.9 M in the optical backbone, 67.3 M in the modality encoders, 8.4 M in the fusion, and the remainder in the neck, the transformer encoder and decoder and the head). All PF-RPN variants were trained at 320 $\times$ 320 pixels with batch size 8, AdamW at 2e-5, weight decay 1e-4, backbone learning rate scaled by 0.5, mixed precision and 8 epochs; the final-epoch checkpoint was used, with no validation-based selection. Two consequences deserve to be stated explicitly. The 900-candidate budget is not an export cap: the decoder carries 900 queries and the test configuration keeps all of them, so a full proposal set is the query set, which is why the proposal route fills the cap on every image while the detector arms do not. And because the proposal route consumes no validation signal while the detector baselines do, the asymmetry works against the proposal route, so the reported contrast is, if anything, conservative. The detector baselines follow the separate recipes given in Section IV-D. Configurations whose comparison required it were run with three random seeds (42, 123, 777). The same schedule and hyperparameters were used across protocols so that fold differences reflect the held-out unit rather than tuning. Because the GPU kernels used in this pipeline are not forced to be deterministic, repeated runs of an identical configuration are not numerically identical. An end-to-end re-run of one held-out fold under the published protocol reproduced the archived average recall to within 0.3 AR points (AR@900 0.2893 to 0.2906; AR@100 and AR@300 differed by 0.0017 and 0.0024). The reproducibility of the training layer is quantified below by adding two seeds to each of three held-out folds; the resulting run-to-run deviations are reported fold-wise rather than as a single constant, because they are not constant across folds (0.008, 0.023 and 0.025 AR against a between-fold standard deviation of 0.159).

D. Baselines

Two detector families served as comparators under identical partitions and preprocessing. Grounding DINO [26] and YOLO-World [27] provided open-vocabulary, text-conditioned baselines, the latter both zero-shot and fine-tuned; YOLOv8s provided a closed-set baseline. Every fine-tuned detector was trained at 320 $\times$ 320 pixels with batch size 16 and initial learning rate 0.01 on the same partitions as the adapted network, for 50 to 100 epochs depending on the arm, and the best validation checkpoint was exported; the zero-shot arms used released weights without further training. All baselines were evaluated with the same proposal-export and scoring procedure as the adapted network, at the same confidence threshold (1e-8) and IoU threshold (0.99), so that the comparison is between paradigms rather than between export settings; because that confidence threshold is already minimal, the number of candidates that an arm emits is its own ceiling rather than a setting of ours.

V. RESULTS

The results are ordered so that the setting comes first and the two transferable findings follow. Sections V-A and V-B fix the measurement: how much recall each paradigm retains on unseen events, and under which candidate budget. Section V-C shows that the size of the paradigm contrast can be forecast without training; V-D and V-E characterise the change modality the criterion is built on; V-F shows that the visibility of the contrast at a practical budget is a property of the verifier; and V-G reports cost.

A. Recall on unseen events separates into three tiers

All external baselines were evaluated on the event-level split, whose four test events are absent from training and which therefore constitutes an unseen-event setting. Every arm was scored through the same export pipeline with a cap of 900 candidates per image. On these unseen events recall separates into three tiers. Zero-shot detectors retain little: Grounding DINO reaches 0.1828 AR@900 and zero-shot YOLO-World 0.1019; the latter produced no proposals at its default threshold and assigned a maximum confidence of 3.6e-4 to any region in this imagery, so its failure is a property of the model on this data rather than of a conservative threshold. Detectors fine-tuned on the five training events recover to about half the recall of the proposal route: 0.3321 for YOLO-World and 0.3203 for the closed-set YOLOv8s. The class-agnostic proposal route reaches 0.7059 on the same export path, and 0.6619 when the same proposals are scored ground-truth weighted; every ratio quoted here uses the image-averaged value, and the weighted one recurs in Section V-F. On this event-level split the two fine-tuned arms land on the same level although one is closed-set and the other text-conditioned, so what separates them from the proposal route here is the paradigm rather than the conditioning; as Section V-B shows, that equivalence does not extend uniformly to the fold protocol. The margin depends on the budget at which it is read: at a strict budget of 100 candidates per image, which every arm can fill, the proposal route stands 7.7-fold above the fine-tuned YOLO-World and 10.1-fold above Grounding DINO, whereas at the 900-candidate cap it narrows to 2.13-fold and 3.86-fold. At the strict budget of 100 candidates the two fine-tuned arms are themselves indistinguishable, at 0.0617 and 0.0606. Earlier measurements of this contrast were taken from exports capped

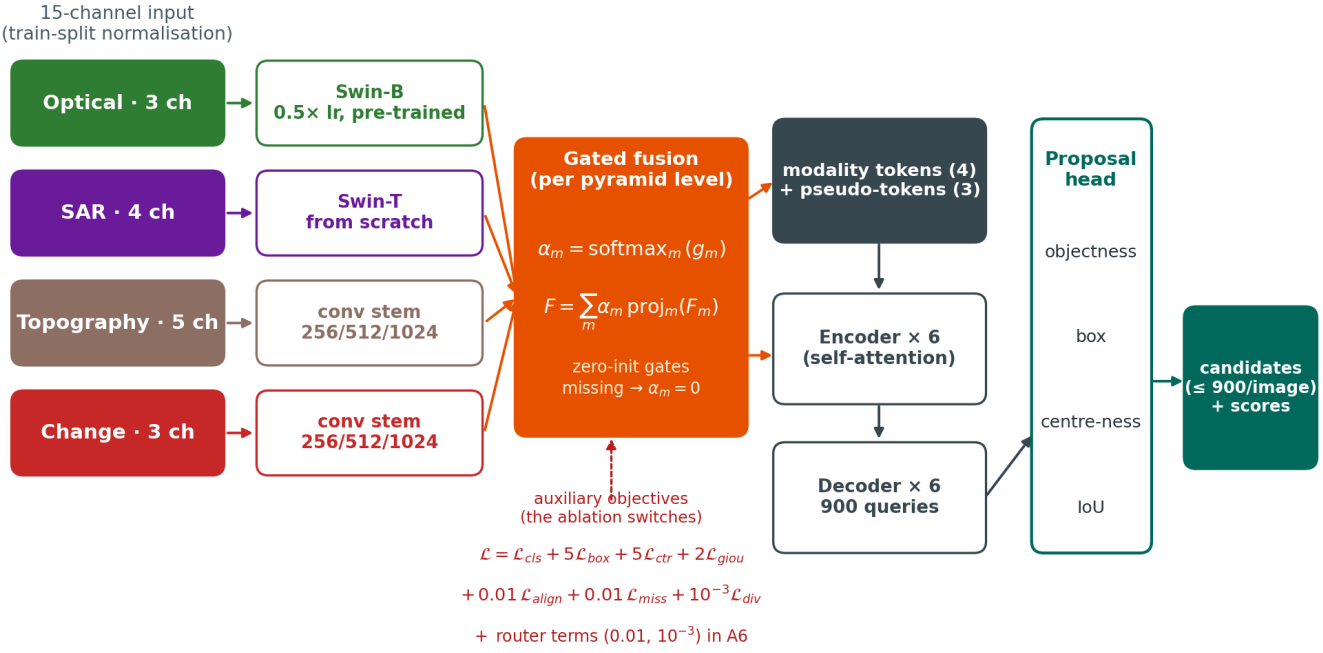


Fig. 2. The proposal route. Each modality is encoded separately (Swin-B for optical, initialised from the released PF-RPN checkpoint; Swin-T for SAR; lightweight convolutional stems for topography and change) and fused at every pyramid level by a gate that is softmaxed over the modalities present in the sample and zero-initialised at the start of training. The fused pyramid and four modality tokens, together with three learnable pseudo-tokens that replace the text prompt, condition a six-layer encoder and a six-layer decoder with 900 queries; the head predicts a class-agnostic objectness, a box, a centre-ness and an IoU for every query. The bracketed auxiliary objectives are the switches varied in the ablation of Table II.

below 100 candidates per image, where both fine-tuned arms returned 0.04 or less. Two causes contribute to the difference from the values above, and we separate them rather than attributing it to the budget alone: re-training the same recipe at that cap raises the closed-set arm from 0.0397 to 0.0617 (+0.022), and the candidate budget accounts for the remainder, 0.0617 at 100 candidates, 0.1364 at 300 and 0.3203 at 900. On the cross-setting folds the closed-set arm was re-exported under the same 900-candidate cap; Section V-B reports that comparison separately because the two protocols have different test-set composition.

B. The prompt-free proposal route retains recall

Across the seven LOEO folds the adapted network reached a mean AR@900 of 0.6749 (s.d. 0.1597) against 0.0030 for the per-fold closed-set control, whose export was capped below 100 candidates per image so that this is not a matched-budget comparison, with seven of seven folds favouring the proposal route.

The candidate counts differ accordingly: on these folds the proposal route emits exactly 900 candidates per image, the fine-tuned YOLO-World a median of 359 to 900, and the closed-set control a median of 118 to 296, so the AR@900 values below are the ceiling each arm reaches from its own candidate stream; the budget-matched comparison is the strict common budget of 100 candidates, given at the end of this subsection. Table I lists the per-fold values for all five arms. Across the three LORO folds it reached 0.7262 (s.d. 0.0395) against 0.0027. Those two ratios are against the weakest comparator in the study, measured under that cap; the matched-budget comparison against the arms that could be re-exported at 900 candidates is given in Section V-A. The two extremes bracket the behaviour honestly: the lowest fold (italy, AR@900 = 0.2896) is the largest event and is dominated by very small objects, while the remaining six folds lie between 0.6907 and 0.7889. Mean LORO recall was not lower than mean LOEO recall even though LORO removes an entire region from training, so survival does not depend on the presence of other events from the same region (Fig. 3 and Fig. 4). The closed-set control plays a different role: its export is capped below 100 candidates per image, so at k=900 it measures what a budget-mismatched comparison would have reported rather than what the paradigm can deliver. On the same folds the two zero-shot detectors, scored through the export pipeline shared by all detector arms, reach mean AR@900 of 0.2524 and 0.1674, while the closed-set control returns 0.0030 from an export capped below 100 candidates per image.

Scored through the same export pipeline as every baseline, the proposal route reaches a ten-fold mean AR@900 of 0.6903, against 0.2222 for fine-tuned YOLO-World, 0.2524 for Grounding DINO zero-shot and 0.1674 for YOLO-World zero-shot; the closed-set arm reaches 0.0978 when allowed up to 900 candidates (its export often produces fewer), but only 0.0433 at the strict common budget of 100 candidates, where fine-tuned YOLO-World reaches 0.0584. Two ratios therefore have to be distinguished. Against fine-tuned YOLO-World the proposal route is 3.11-fold higher at 900 candidates; against Grounding DINO zero-shot, which is the strongest external

arm on this ten-fold protocol, it is 2.73-fold higher. At the strict common budget of 100 candidates the proposal route is 7.94-fold above fine-tuned YOLO-World, so the contrast shrinks monotonically as the budget grows (7.94-fold at 100, 3.11-fold at 900). The two fine-tuned arms are not equivalent across folds: at 100 candidates fine-tuned YOLO-World is 1.35-fold above the closed-set arm, and eight of the ten folds place the closed-set arm below it, although the two remain within the same order of magnitude. The equivalence seen on the event-level split, where both are scored at the same 900-candidate cap, therefore does not carry over to the fold protocol, and the proposal route remains above both. The two orders of magnitude that the earlier unequal budget and export cap produced were a property of the measurement conditions, not of the paradigm.

Two features of the pipeline set the scale against which these differences should be read. The input data reproduce exactly: the delivered tree was re-derived from the public source with the original converter and is bit-identical for all 10,015 tiles and all 60,090 arrays, every split and ground-truth manifest references zero missing tiles, and the training-set normalisation statistics recomputed identically (6,813 tiles, 111,624,192 pixels, all 15 channel means and standard deviations). The training protocol, by contrast, is not deterministic. A complete re-run of one held-out fold reproduced the archived AR@900 to +0.0013, and adding two seeds to each of three folds gives fold-wise maximum deviations from the same-configuration re-run of 0.0076 AR (the lowest fold, italy), 0.0234 AR (chimanimani) and 0.0251 AR (the highest fold, indonesia), against a between-fold standard deviation of 0.159. Run-to-run variability therefore stays more than six times below the between-fold dispersion, and the envelope is a fold-wise range rather than a constant. Against the largest observed deviation of 0.025 AR the two main effects remain more than an order of magnitude larger: the four-modality gain is 0.36 AR (2.03-fold, 14 times), and the paradigm contrast is 0.67 AR for leave-one-event-out and 0.72 AR for leave-one-region-out (27 and 29 times). Single-fold contrasts the size of the change-modality contribution, which spans 0.031 to 0.30 AR, are not all resolvable from run-to-run noise on one seed, and the smallest was therefore adjudicated by a paired same-seed re-run: across three seeds the change-modality gain is 0.024 ± 0.006 AR, with all three paired differences positive (Supplementary S1). No conclusion in this study rests on a difference smaller than the run-to-run variation of the comparison it is drawn from.

Realised candidate count per image: 900 for the proposal route and for Grounding DINO on every image, a median of 359-762 for fine-tuned YOLO-World (it fills the cap on one fold) and 118-296 for YOLOv8s; the medians are estimated on a 150-image sample per fold, so AR@900 is the ceiling each arm reaches from its own pool.

C. A training-free criterion forecasts the paradigm gap

The choice between a proposal front end and a semantic detector matters most before any model is trained, so we asked whether the size of the gap can be anticipated from the data alone. We used the separability of the change signal, which requires no model and no training, only the change stacks and the available masks of the event under consideration. The statistic is defined as follows. For a tile, —Δ— is the mean of the absolute values of the three change channels (dNDVI, dNDWI, dNBR), evaluated pixel by pixel. Within the tile, the masked pixels give one aggregate (their mean —Δ—) and the remaining pixels give the background aggregate (their mean —Δ—); a tile without ground truth contributes its whole-tile mean to a separate negative-tile pool. The AUC is computed on these per-tile aggregates rather than on individual pixels, and it therefore asks whether tiles that contain a landslide carry stronger change than their own backgrounds. A fold is summarised by the same computation over all of its tiles, so the per-fold values of Fig. 5 and the mean AUC of Section V-E are the same quantity on the same convention. Because the statistic is built from pixel-level masks and then aggregated per tile, we call it pixel-level separability throughout, to keep it distinct from tile-level appearance statistics. Across the ten held-out folds this statistic is strongly associated with the paradigm gap (proposal route over Grounding DINO zero-shot): Pearson r = +0.915, Spearman ρ = +0.758, bootstrap 95% CI [+0.682, +0.986]. Every interval quoted in this subsection is a percentile bootstrap that resamples folds; Fisher-z intervals are not used, because with ten folds a single influential fold makes the sampling distribution of r markedly skewed, and a percentile interval reflects that skew where a Fisher-z interval does not. The association is negative and comparably strong for the detectors taken individually (Grounding DINO zero-shot, r = -0.924; YOLO-World zero-shot, r = -0.906), so the gap opens where the change signal is more separable, an interpretation consistent with the detector degrading there while the proposal route does not; that reading follows from the same ten-fold correlations and is not independently tested. The statistic predicts the contrast against YOLO-World zero-shot less tightly (r = +0.790, CI [+0.401, +0.959]). Crucially, it shows no detectable association with the absolute recall of the proposal route (r = -0.266, CI [-0.670, +0.823], which crosses zero): at ten folds the statistic forecasts the contrast between paradigms rather than the level of any single model (Fig. 5). That interval is wide and asymmetric because the association is not robust to one fold: dropping italy, the fold with the lowest proposal-route recall, moves r to +0.44. The claim we rely on is therefore the sign of the association with the gap, not the magnitude of the association with absolute recall.

We verified that the relationship is not carried by one fold. Fitting the relation on nine folds and predicting the tenth gives a predicted-versus-observed correlation of 0.856 (Spearman 0.661, RMSE 0.789) with the correct sign in all ten folds, and deleting any single fold changes the full-sample correlation by at most 0.040. The diagnostic therefore transfers across events and regions within this benchmark. (see S1-D).

The separation is sharp enough to serve as a two-regime description of these ten folds. Where separability exceeds 0.70 the detector recovers 17.7-26.9% of the proposal route's recall (gap 3.71-5.65; 22% of the route's recall at the group mean), and where it falls below 0.50 it recovers 63.7-72.8% (gap 1.37-1.57; 67% at the group mean); no fold lies between 0.492

TABLE I
PER-FOLD AVERAGE RECALL AT 900 PROPOSALS ON THE TEN CROSS-SETTING FOLDS (IMAGE-AVERAGED).

Fold	PF-RPN	FT-YOLO-World	YOLOv8s	Grounding DINO	YOLO-World ZS
Italy (LOEO)	0.2896	0.1051	0.035	0.0515	0.0337
Kyrgyzstan1 (LOEO)	0.7559	0.2694	0.1399	0.4835	0.3441
New Zealand (LOEO)	0.6907	0.443	0.1604	0.5031	0.3166
Chimanimani (LOEO)	0.735	0.154	0.0532	0.155	0.092
Hiroshima (LOEO)	0.7315	0.1779	0.0737	0.1481	0.0963
Indonesia (LOEO)	0.7889	0.2502	0.0908	0.1675	0.0772
Hokkaido (LOEO)	0.733	0.2105	0.0972	0.1981	0.1832
East Asia (LORO)	0.7307	0.2001	0.0877	0.1937	0.1405
Central Asia (LORO)	0.6757	0.2487	0.1207	0.4443	0.3055
South-East Asia (LORO)	0.7721	0.1625	0.1192	0.1795	0.0845
Mean	0.6903	0.2222	0.0978	0.2524	0.1674
s.d.	0.1373	0.0877	0.0365	0.1526	0.1082

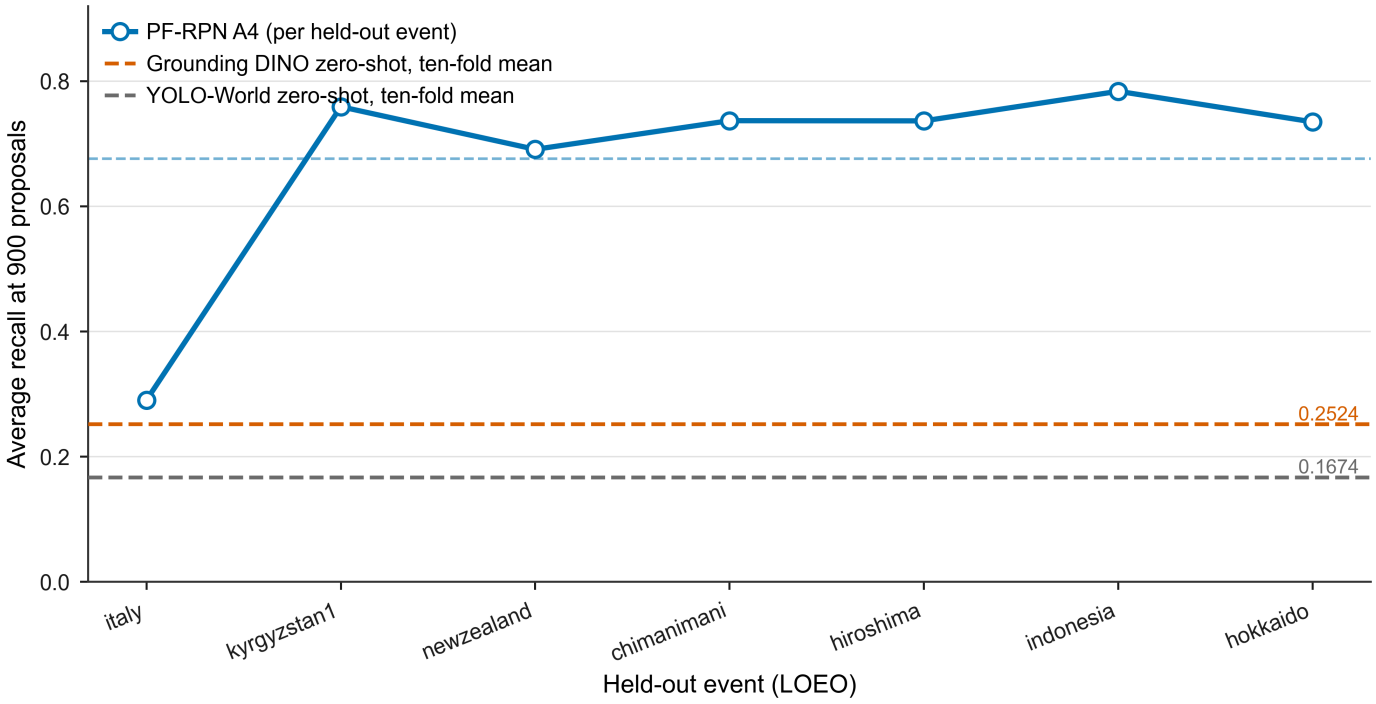


Fig. 3. Cross-event recall. AR@900 for the prompt-free proposal route on each of the seven held-out events, with the ten-fold mean of the two zero-shot detectors marked as reference lines. The closed-set control is not plotted: its export was capped below 100 candidates per image.

and 0.705 on the separability axis. Two limits apply to that description. The cut is a property of these ten folds rather than a calibrated rule, because the threshold was not fixed in advance; and a threshold fitted on nine folds and applied to the tenth classifies all ten folds correctly (gap above or below 2.5) only because the two groups differ by 0.213 in AUC, so that check establishes robustness to the choice of fold rather than independent calibration. A calibrated rule would need a second benchmark; what transfers prospectively is the ordering, as the leave-one-fold-out check above shows.

Its scope must be stated precisely. It is training-free, not annotation-free: the statistic is computed from the change stack together with the landslide masks of the event being characterised, so it applies where a reference inventory already exists for the region or an initial delineation has been made. It therefore answers the routing question as it arises in practice, where a reference inventory exists for the region or an initial delineation has been made. The fully unannotated event is a different problem — placing the first landslide of an unscreened event into a candidate set — and that problem is addressed by the interval result of Section V-B, not by a mask-based statistic. Outside that scope the choice of paradigm remains open; within it, the statistic is a usable screening signal. Training-free separability measures have been used to predict transfer performance [39]–[41]; the quantity used here serves a different prediction target (the choice of detection paradigm in landslide candidate discovery) and is defined on the raw change channels, so it depends on no pre-trained model.

D. Four-modality adaptation, the role of event change, and the interval it depends on

The event-level ablation over ten modality configurations (A0 optical only; A1 = optical + DEM; A2 = optical + SAR; A3 = optical + SAR + DEM; A4 = A3 + event change; A5 = A4 + terrain-aware token; A6 = A5 + cross-modal router; A7 =

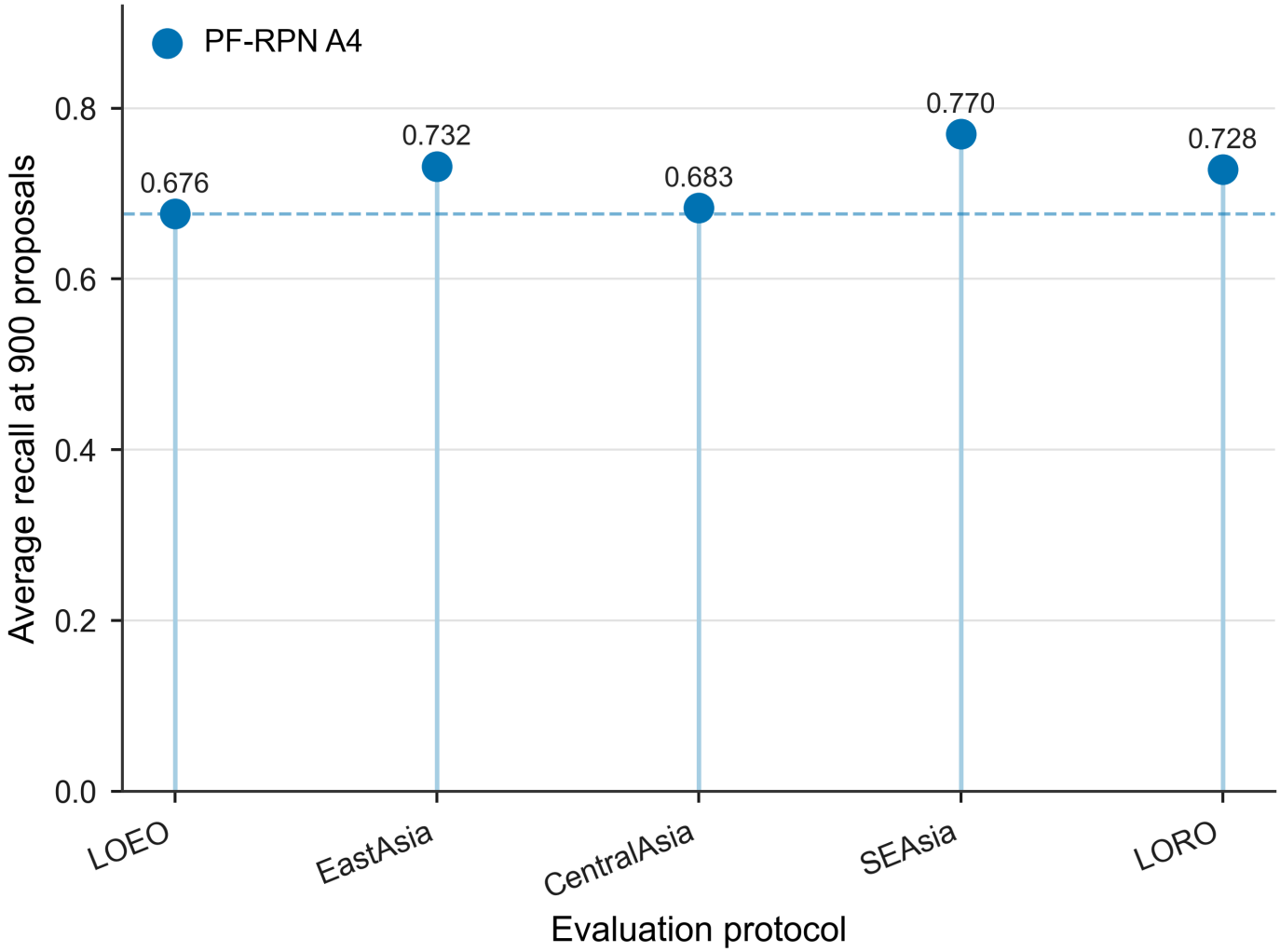


Fig. 4. Cross-region transfer. AR@900 on each of the three held-out regions, shown against the mean of the seven held-out events.

A6 without the router; A8 = A6 without SAR; A9 = A6 without the change channels) shows that accuracy is concentrated in the temporal-change modality. An optical-only baseline (A0) reached AR@900 = 0.3533; adding SAR and topography (A3) raised this to 0.4574; adding the event-change channels (A4) raised it to 0.7167 — a 2.03-fold gain over A0 and a 1.57-fold gain over A3. The figures above are from a single seed; across three random seeds the means are 0.4642 for A3 and 0.7127 for A4, a difference of +0.2485 that survived paired testing with Benjamini-Hochberg correction (q = 0.00075). Table II and the permutation control of Fig. 7 use the A3 single-seed value, whereas the change-modality comparison of Section V-E uses the three-seed mean. Structural additions behaved differently. A cross-modal router (A6, 0.7059) is indistinguishable from A4 (0.7167): the difference of 0.011 is the same-configuration run-to-run difference measured in Section V-B, so A6 brings no gain but is not a degradation either. A terrain-aware token (A5, 0.6905) sits at the edge of that envelope (0.026 against a maximum of 0.0251) and adds no consistent gain. Neither is folded into the contribution. A7 (0.7029) differs from A6 by 0.003 and is not separable from it at this seed count. Recall on small objects (area below 32 $\times$ 32 pixels at the 128-pixel tile scale, the small bucket of the evaluator) was markedly higher for the adapted network (0.6117) than for any baseline (at most 0.0685) on the same event-level test split. The advantage is visible at every proposal budget: at 100 proposals the adapted network reached AR = 0.4882 against 0.0962 for A0, and at 300 proposals 0.6171 against 0.1974 — all ablation-run values; the budget analysis of Section V-G uses the main-run curve, which gives 0.4652 and 0.5908 at the same two budgets. The full modality ablation is given in Table II. Table II reports one training run per configuration. The main results of Sections V-A, V-G and VI-A come from a separate, independent training run of the A4 configuration, which is also the checkpoint frozen for the interventions of Fig. 7: it reaches 0.7059 at k = 900, 0.5908 at 300 and 0.4652 at 100, against 0.7167, 0.6171 and 0.4882 for the A4 row of Table II. The value 0.7059 also appears in the A6 row, but the two runs are unrelated: their k = 900 values differ in the fifth decimal (0.70589 against 0.70592) and their k = 300 values differ clearly (0.5908 against 0.6000), so the coincidence is numerical rather than a mislabelling.

One training run per configuration. The main results of Sections V-A, V-G and VI-A use a separate, independent run

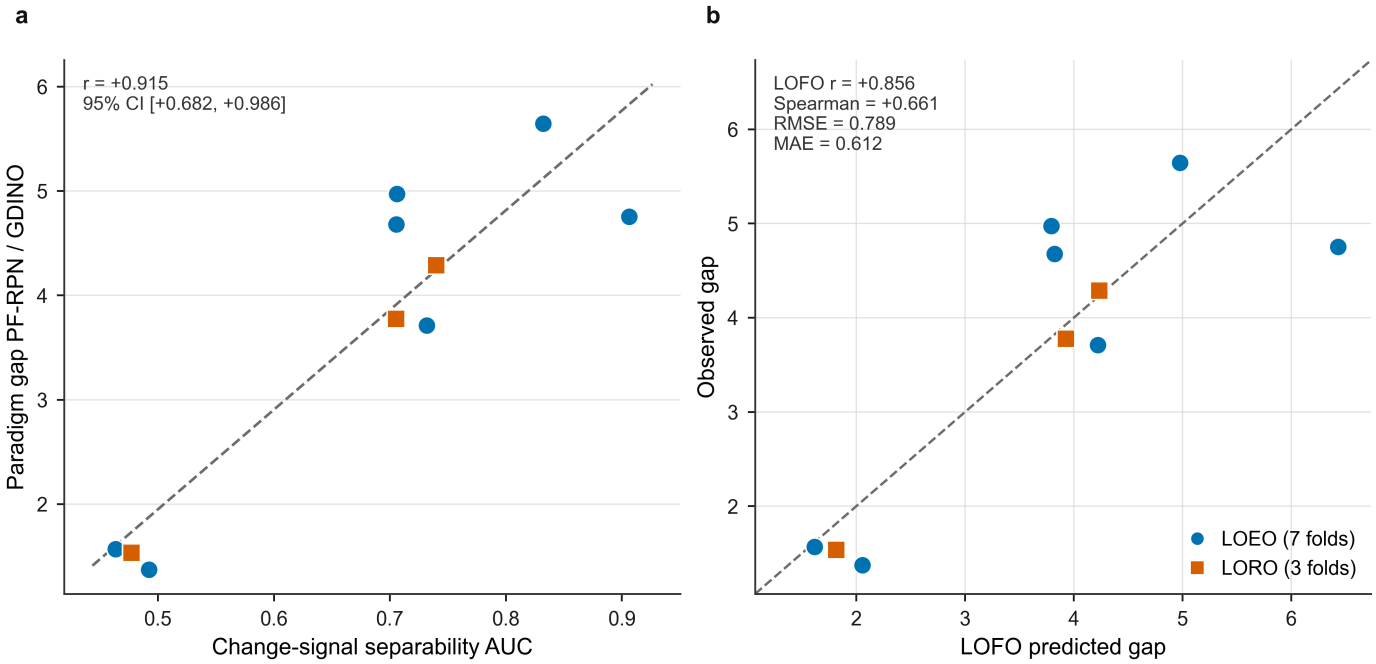


Fig. 5. Separability of the change signal against the paradigm gap on the ten held-out folds. Left: separability AUC against the per-fold gap between the proposal route and Grounding DINO zero-shot. Right: gap predicted under leave-one-fold-out against the observed gap.

TABLE II
MODALITY ABLATION A0-A9 ON THE EVENT-LEVEL SPLIT; THE THREE BUDGETS ARE K = 100, 300 AND 900 PROPOSALS PER IMAGE.

ID	Input configuration	AR@100	AR@300	AR@900
A0	Optical only	0.0962	0.1974	0.3533
A1	Optical + DEM	0.1426	0.2527	0.4220
A2	Optical + SAR	0.1439	0.2622	0.3745
A3	Optical + SAR + DEM	0.1590	0.2878	0.4574
A4	A3 + event change	0.4882	0.6171	0.7167
A5	A4 + terrain-aware token	0.4678	0.5859	0.6905
A6	A5 + cross-modal router	0.4797	0.6000	0.7059
A7	A6 without router	0.4797	0.5998	0.7029
A8	A6 without SAR	0.4712	0.5858	0.6729
A9	A6 without change	0.1652	0.3048	0.4799

of the A4 configuration (0.7059 / 0.5908 / 0.4652 at k = 900 / 300 / 100); the 0.7059 of the A6 row is a different run that coincides at four decimals only (0.70592 against 0.70589). Single-seed throughout; the three-seed means are 0.4642 for A3 and 0.7127 for A4 (paired test q = 0.00075). For this revision, A0 and A9 were re-run with the same recipe, that is batch size 4, eight epochs and seed 42. A0 gives 0.3714 with a 95% confidence interval of [0.350, 0.392], which lies within the run-to-run band of the archived value, while A9 gives 0.4337, which is 0.046 below its archived value and is reported as an open discrepancy in the supplementary material. (see S1-B).

To test whether this gain reflects the information carried by the change channels rather than the three additional input channels themselves, we intervened on the trained network at evaluation time without retraining. Permuting the spatial content of the three change channels while leaving the channel count, tensor shape and network unchanged lowered AR@900 from 0.7059 to 0.3866 on the same 1,759-image test set and the same frozen checkpoint used for the per-channel interventions; applying the identical operation to SAR cost 0.015 and to topography 0.007, so the loss is attributable to the change information rather than to the perturbation itself, and it lands near the model trained from the outset without change information (A3, 0.4642). Zeroing individual channels produced intermediate losses: 0.6087, 0.5577 and 0.7090 for the first, second and third change channel (dNDVI, dNDWI and dNBR) respectively, so the water-index difference carries most of the signal, the vegetation-index difference contributes substantially, and the burn-severity difference contributes little on this benchmark. The advantage therefore rests on the information carried by the change channels rather than on added capacity or on the number of input modalities.

The change channels carry the largest single-modality contribution (Section V.D). A natural question is whether their value depends on which two acquisitions are differenced, how that interval should be chosen, and where in the pipeline it must be applied. We answer in three steps, each of which changes one thing at a time.

Fig. 6 separates two conventions that are usually conflated. Conventional pipelines difference the same interval during training and at inference, so the interval never presents itself as

a choice. The scheme tested here anchors the training window on the event and leaves the deployed window at the long baseline, which turns the interval into a training-time decision rather than an input-time constant.

First, the interval is determined by the calendar. Section III.B shows that the annotated window is recovered exactly in 97.9% of 3,135 tiles from the event date and the tile's own acquisition calendar. Two structural facts follow. The window is a single adjacent acquisition pair, so its span is one acquisition rather than a fraction of the series, and it sits near the middle of the series, whereas the deployed long baseline spans 678 days. Interval selection is therefore a position problem rather than a span-tuning problem, and position is a calendar quantity.

Second, pixel evidence favours the correct interval but not the shortest one. On four test events and roughly 400 annotated tiles, with the official masks as ground truth, the event window widened by one acquisition on each side separates landslide pixels better than the long baseline: AUC 0.848 versus 0.777, and recall at a 1% false-positive rate 0.270 versus 0.195, with a paired difference of +0.066 (95% CI [+0.046, +0.060]). The single adjacent pair is worse than the long baseline (0.744 and 0.119, paired difference -0.045, CI [-0.071, -0.055]), because only 77.6% of its pixels are usable in both frames, against 98.5% for the widened window. A window of the same span placed away from the event degrades to 0.506 and 0.016. (see S1-F1).

Third, the interval must be committed during training, but the deployment input need not change. The post-hoc route fails. With the same machine and the same checkpoint, replacing the change channels of a model trained on the long baseline by the widened event stack lowers AR@900 from 0.7085 to 0.6217, and by the single-pair stack to 0.5460; Fig. 7 places every interval arm, training-side and inference-side, on one axis. Re-normalising with the variant's own statistics lowers it further, to 0.5941 and 0.5119, so the drop reflects a content dependency in the weights rather than a scale mismatch. Training with the correct interval is a different matter. Fixing deployment to the long baseline, the training interval orders performance monotonically: 0.6969 for the single pair, 0.7094 for the long baseline, and 0.7474 for the widened event window over three seeds (Table III). The gain is not obtained by changing what the model sees at run time, but by changing what it learns from. (see S1-E3).

The gain is stable across training seeds. Averaged over three seeds, AR@100 rises from 0.4652 to 0.5197 with a standard deviation of 0.0091 across seeds, AR@300 from 0.5886 to 0.6488 with a standard deviation of 0.0054, and AR@900 from 0.7094 to 0.7474 with a standard deviation of 0.0105. Combining the target's seed variance with the anchor's run-to-run variance of 0.0059, taken from three archived runs of the same configuration, puts the AR@900 gain at 5.5 standard errors, the AR@300 gain at 13.0 and the AR@100 gain at 8.7. A paired per-image bootstrap provides an independent second estimate: run on the first seed and the 673 test images that carry ground truth, it gives +0.0468 at k=900 with a 95% confidence interval of [+0.029, +0.065] and a sign test with p below 1e-7. The two estimates differ in basis by design — the seed average is taken over all 1,759 test images, the bootstrap over the paired 673 — and they agree in sign and in order of magnitude, so the gain survives both aggregations. On that seed the gain holds in all three size buckets (small 0.607 to 0.661, medium 0.846 to 0.880, large 0.885 to 0.904) and in three of the four events, and micro recall at 900 rises from 0.660 to 0.712 (Table IV; see S1-I).

Two readings follow. The intervention is free at run time, because the deployed change stack is unchanged, so the cost model of Section V-G is untouched. And the gain is largest where candidates are scarce, which is the regime that matters when proposals are consumed by a fixed-budget downstream stage. Both readings are made under the event-level protocol; Section VI-E reports a cross-protocol fold that does not reproduce the gain.

The channel count is not the explanation. To separate the value of the event interval from the value of carrying more change channels, we trained the same network on a 6-channel change stack in which the three additional channels repeat the deployed long-baseline stack. That arm reaches AR@900 0.7041 (95% CI [0.6822, 0.7246]) against 0.7094 for the 3-channel anchor, so three additional channels that carry the long interval again contribute nothing. Giving the same three additional channels the event window, so that both intervals enter the model, reaches 0.7583 over two seeds (0.7634 and 0.7532, s.d. 0.0072), and its confidence interval does not overlap the control's. The gain of that arm is therefore attributable to the interval it carries rather than to the width of the input, and it is concentrated where candidates are scarce: against the zero-cost arm of the same seed, a paired per-image bootstrap gives +0.049 at k=100 (95% CI [+0.025, +0.073]) and +0.029 at k=300 ([+0.008, +0.050]), but +0.007 at k=900 ([-0.011, +0.026]). This configuration needs the event stream at inference as well, so we report it as an optional extension of the free intervention rather than as part of it (Table III; see Table S1-E5).

E. The benefit of event change follows the event; separability forecasts its sign, not its size

Removing the change modality and retraining (A3) in every cross-setting fold isolated its contribution directly. Across ten folds the change modality contributed between 1.05-fold and 1.93-fold (mean 1.36-fold), so its value is a property of the event rather than a constant of the method. We then asked whether that value can be anticipated without training, using the same separability statistic. In the manner of a correlation-based mechanism test, we related that quantity to the observed benefit. The association was moderate (Pearson r = +0.722 across ten folds), but separability is far from uniformly informative: across the ten folds the mean AUC was only 0.676 and three folds were close to chance (0.463, 0.492 and 0.477). The residuals are informative rather than noise around a tight line: kyrgyzstan1 has the lowest separability (AUC 0.463) but retained a 1.07-fold benefit, and the ordering does not follow separability: hokkaido (AUC 0.732) gained only 1.14-fold where hiroshima (AUC 0.706) gained 1.67-fold, and the largest single-fold benefit (italy, 1.93-fold) sits at high separability

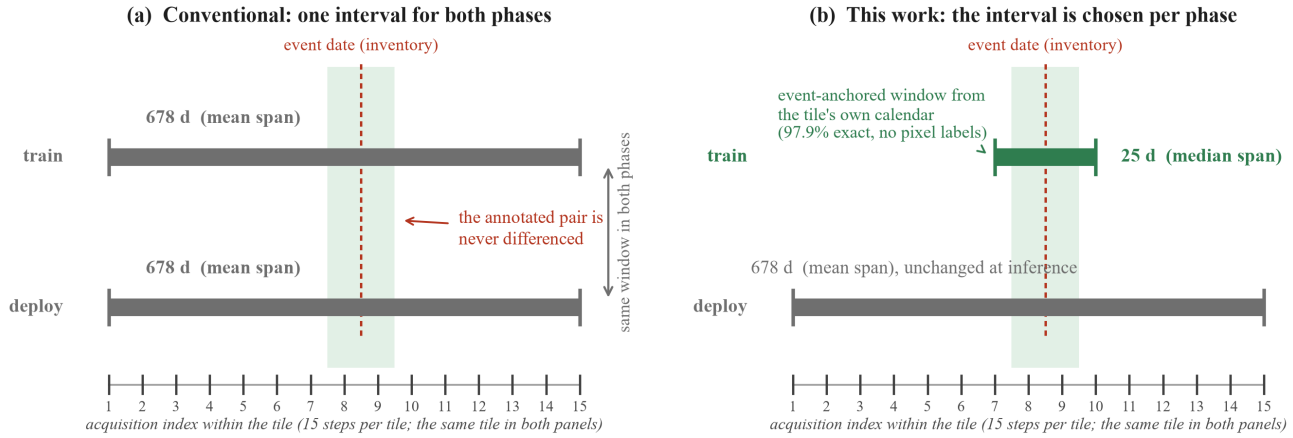


Fig. 6. The interval assumption and its removal. (a) The conventional pipeline differences each tile between its first and last acquisition (678 d on average) during training and again at inference, so the interval is silently treated as a constant and the annotated pair sitting in the middle of the series is never used. (b) The scheme tested here trains the change channels on a window anchored on the event, the annotated pair widened by one acquisition on each side (median 25 d, recovered from the tile's own calendar in 97.9% of 3,135 tiles without pixel annotation), and leaves the deployed input at the long baseline, so the inference input, cost and code are unchanged. Giving the event window to a frozen long-baseline model lowers AR@900 from 0.7085 to 0.6217, which is why the window has to be committed during training. If the event calendar is also available at deployment, a second change stream carrying the event window can be added alongside the baseline; that configuration reaches 0.758 over two seeds, but it needs the extra stream at inference and is therefore not part of the free intervention.

TABLE III

TRAINING INTERVAL AGAINST DEPLOYMENT INTERVAL (AR@900, FOUR-EVENT TEST SPLIT, 1,759 IMAGES, 4,227 GROUND-TRUTH BOXES). EVERY ROW IS SCORED BY THE SHARED EXPORT ON ONE MACHINE WHOSE ANCHOR IS 0.7094; ROWS THAT AVERAGE SEVERAL SEEDS SAY SO EXPLICITLY, AND THE SINGLE-SEED AND SINGLE-PAIR ROWS ARE SINGLE RUNS. †THE TWO FROZEN ROWS REPLACE ONLY THE CHANGE STACK OF THE FROZEN ANCHOR MODEL, ON A SECOND MACHINE WHOSE OWN ANCHOR IS 0.7085, SO THEIR DIFFERENCE IS COMPUTED WITHIN THAT BLOCK. THE TWO SIX-CHANNEL ROWS CARRY THREE ADDITIONAL CHANGE CHANNELS: IN THE CONTROL THOSE CHANNELS REPEAT THE LONG-BASELINE STACK, AND IN THE LAST ROW THEY CARRY THE EVENT WINDOW, SO THAT BOTH INTERVALS ENTER THE MODEL.

Training window	Deployment window	AR@900	Difference
long baseline	long baseline	0.7094	anchor
long baseline, stack duplicated to 6 ch	long baseline (6 ch)	0.7041	-0.005
single adjacent pair	long baseline	0.6969	-0.013
widened event window (3 seeds)	long baseline	0.7474	+0.038
long + widened event (6 ch, 2 seeds)	long + widened event (6 ch)	0.7583	+0.049
single adjacent pair	single adjacent pair	0.5783	-0.131
widened event window (1 seed)	widened event window	0.7179	+0.009
long baseline (frozen)	widened event window	0.6217†	-0.087

TABLE IV

BUDGET DECOMPOSITION OF THE WIDENED-EVENT TRAINING ARM, PAIRED OVER THE 673 TEST IMAGES WITH GROUND TRUTH. REL. IS THE RELATIVE CHANGE AND S.D. THE SEED STANDARD DEVIATION; THE SAME PAIRING IS USED IN ALL COLUMNS.

k	anchor	widened	Δ	rel.	s.d.
100	0.4652	0.5197	+0.0545	+11.7%	0.0091
300	0.5886	0.6488	+0.0602	+10.2%	0.0054
900	0.7094	0.7474	+0.0380	+5.4%	0.0105

(AUC 0.832) without carrying the rest of the relationship. We verified that the pattern is not a mask-alignment artefact. The stored pixel mask is the union of the instances in a tile, and the bounding box of that union coincides exactly with the box enclosing all annotations in 92.5% of tiles (median intersection-over-union 1.000, mean 0.988 over 800 tiles from the eight folds; the residual cases are tiles in which a single annotation is small relative to the union and the two boxes differ). The correspondence holds for the tile-level union, not for individual instances: against the best-matching single annotation the mask-derived box reaches a mean intersection-over-union of only 0.230, so the statistic is a property of the tile rather than of a particular landslide. Following the convention of such mechanism checks, we therefore report the result as a partial explanation rather than a mechanism: separability sets the direction and the size class of the routing decision across the ten folds, while the magnitude of the change-modality benefit carries an event-specific component that it does not encode (Fig. 8). The association describes how much the change modality contributes to a given event; it does not describe how far apart the two paradigms lie. That contrast, and the question of whether it can be anticipated at all, is the subject of the previous subsection. The quantities above are fold-level aggregates, and the analysis of this subsection rests on them; Fig. 9 shows what the contrast looks like on individual tiles.

F. A verification stage: the paradigm gap depends on the verifier

Candidate discovery only becomes useful if the candidate set can be reduced to a shortlist, so we added a verification

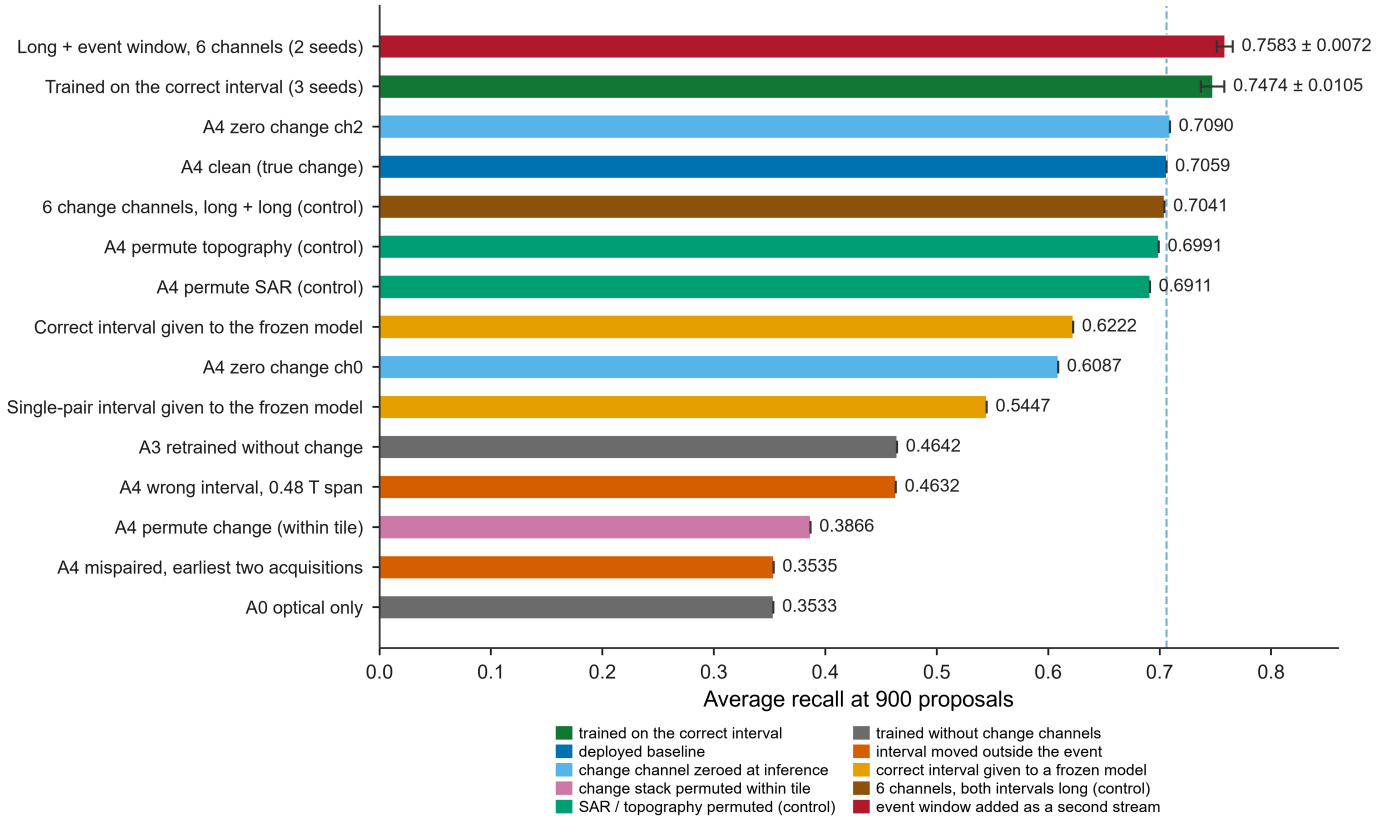


Fig. 7. The event interval changes recall only if it is committed during training. AR@900 on the event-level test split for the adapted network, sorted by value. From the bottom up: the model retrained without change information and the optical-only baseline; the change stack recomputed outside the event window, at the earliest two acquisitions and at the longest span the archive allows (0.48 of the true span); the change stack permuted inside each tile, with SAR and topography permuted as controls for the permutation itself; the change channels zeroed individually at inference; the correct event interval and the single adjacent pair given to the frozen model; the deployed baseline; and, at the top, the models retrained with an event-anchored interval. Two of those arms carry six change channels, so that the extra channels can be varied independently of the interval: repeating the deployed stack leaves recall on the baseline (0.7041 against 0.7094 on this machine), whereas giving the extra channels the event window, and thus both intervals at once, reaches 0.7583 ± 0.0072 over two seeds. Giving the correct interval to a frozen model costs 0.087 recall, whereas training on it gains 0.038. The six-channel arms and the frozen substitutions were run on one machine whose own baseline is 0.7094, 0.005 above the archived 0.7059 marked by the dashed line, which is inside the run-to-run noise of 0.0059; the remaining bars are single archived runs.

stage that re-ranks the candidate pools produced by the shared export on the event-level test split (1,759 images, 4,227 ground-truth boxes). Each pool is capped at 900 candidates per image: the proposal route and Grounding DINO zero-shot fill that cap on every image, whereas the zero-shot YOLO-World pool does not (median 847 candidates per image). The three arms of this stage are the proposal route, Grounding DINO zero-shot and YOLO-World zero-shot, and each is scored on the pool it actually produces. To keep the paradigms comparable the verification stage is shared: one scorer is trained and then applied unchanged to the candidate pools of all three arms, and its inputs are restricted to evidence inside the proposed box (the change response, terrain attributes and the optical crop) so that no detector's own scores enter the comparison.

Six scorers, none of which uses a detector's own scores, separate the arms very differently at a practical budget. Ranking candidates by the mean in-box magnitude of the change signal alone gave, at ten proposals per image, a recall of 1.85% for the proposal route against 1.66% for Grounding DINO zero-shot, a separation of 1.11-fold; a logistic verifier trained on eleven box-interior statistics narrowed this to 1.05-fold (2.37% and 2.25%); a verifier built on a frozen vision-language embedding of the in-box optical crop widened it to 1.80-fold (2.89% and 1.61%); and the same embedding trained with a listwise ranking loss over the candidates of one image, rather than pointwise classification, reached 3.31%, a separation of 1.92-fold. Recall at 900 proposals is identical for every scorer, as it must be, and on the ground-truth-weighted convention it remains 4.15-fold apart (0.6619 against 0.1595 for the proposal route and Grounding DINO zero-shot); the image-averaged convention used elsewhere in this paper gives 3.86-fold for the same comparison (0.7059 against 0.1828). Table V summarises the six scorers.

Recall is ground-truth weighted, as in Section V-F. For reference, the proposal route scoring its own candidates with its native confidence reaches 22.65% at the same budget.

Two controls show that this widening is a property of the scorer and not an artefact of how it is trained. First, the natural training mixture is strongly unbalanced (the proposal route supplies 72% of the positive examples) so we retrained with every arm contributing the same number of positives and negatives; the separation at ten proposals rose from 1.80-fold to 2.42-fold (3.10% and 1.28%). Second, we left one

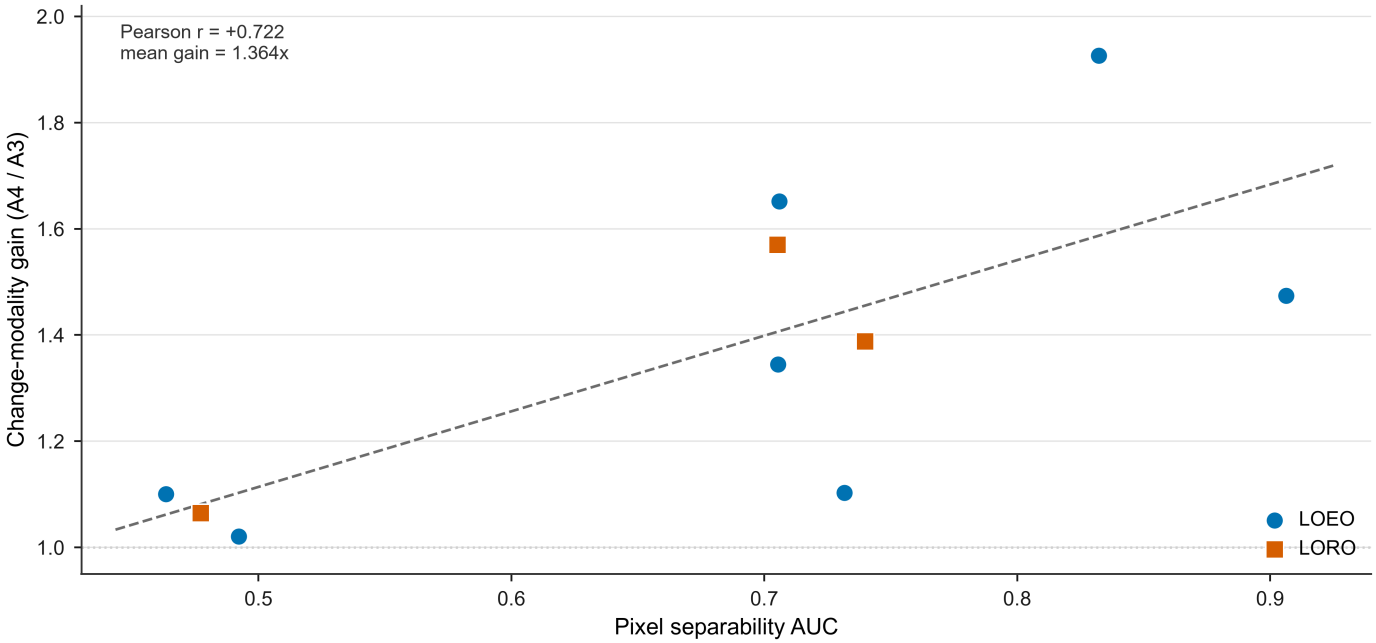


Fig. 8. Change-modality benefit versus change separability (delta AR@900 against the per-tile separability AUC defined in Section V-C) for the ten held-out folds.

TABLE V
SHARED-VERIFIER ABLATION AT TEN PROPOSALS PER IMAGE ON THE EVENT-LEVEL TEST SPLIT; THE LAST COLUMN IS THE SEPARATION BETWEEN THE PROPOSAL ROUTE AND GROUNDING DINO ZERO-SHOT.

Shared scorer (k = 10)	PF-RPN (%)	Grounding DINO (%)	YOLO-World ZS (%)	PF-RPN / GDINO
Untrained: in-box change magnitude	1.85	1.66	1.37	1.11x
11 box-interior statistics + pointwise LR	2.37	2.25	1.82	1.05x
Frozen CLIP embedding + pointwise LR	2.89	1.61	1.25	1.80x
Frozen CLIP + pointwise LR, arms balanced	3.10	1.28	0.88	2.42x
Frozen CLIP, leave-one-arm-out	3.31	1.30	0.90	2.55x
Frozen CLIP + listwise ranking (selected)	3.31	1.73	1.66	1.92x

arm out of training and evaluated the verifier on that arm alone: the held-out arm scored as well as it did in the balanced model (leave-one-arm-out 3.31% against 3.10% in the balanced model for the proposal route; 1.30% against 1.28%; 0.90% against 0.88%), and within a verifier that never saw the proposal route the proposal route still outscored an arm that was in its training set, giving a cross-arm separation of 2.55-fold. The shared scorer therefore recognises landslides inside the box rather than the signature of any particular detector.

Comparing the six scorers separates two levers. Changing the features, from eleven box-interior statistics to a frozen vision-language embedding, raised the recall of the proposal route at ten proposals by 22% (2.37% to 2.89%); changing the objective, from pointwise classification to a listwise ranking loss over the candidates of one image, raised it by a further 15% (2.89% to 3.31%). The two levers are therefore comparable in size, and together they move the same candidate pool from 1.85% to 3.31%, a factor of 1.8. The number of training epochs was chosen on a held-out validation split of 300 images drawn from the training events, and the test split was evaluated once; the validation curve peaks at twelve epochs (.0483) and declines thereafter (.0435 at forty), a broad plateau rather than a sharp optimum, so the configuration is not limited by insufficient training. Repeating the epoch selection under an event-level holdout, with the whole hiroshima event (161 images) excluded from training instead of 20% of the images, changed the selected epoch from twelve to twenty-eight while leaving the test ratio unchanged within 0.03 fold (1.89 against 1.92) and AR@900 bit-identical, so the verifier comparison does not depend on sharing events between selection and training.

The verification stage therefore determines whether the paradigm contrast is visible at a practical budget at all. Every scorer with semantic content — the frozen embedding under either objective, with balanced training, and under leave-one-arm-out evaluation — places the separation at ten proposals between 1.80-fold and 2.55-fold, whereas the two statistics-based scorers place it between 1.05-fold and 1.11-fold. The conclusion therefore holds under two training objectives, two training-set constructions, a leave-one-arm-out protocol and two training schedules (1.92-fold and 2.01-fold). The absolute level nevertheless remains low: the best scorer recovers 3.31% of landslides within ten proposals per image, about three times chance, against 22.65% for the proposal route scoring its own candidates with its native confidence at the same budget. That last figure is exactly why native confidence cannot be used in a cross-paradigm comparison: it would credit the model that

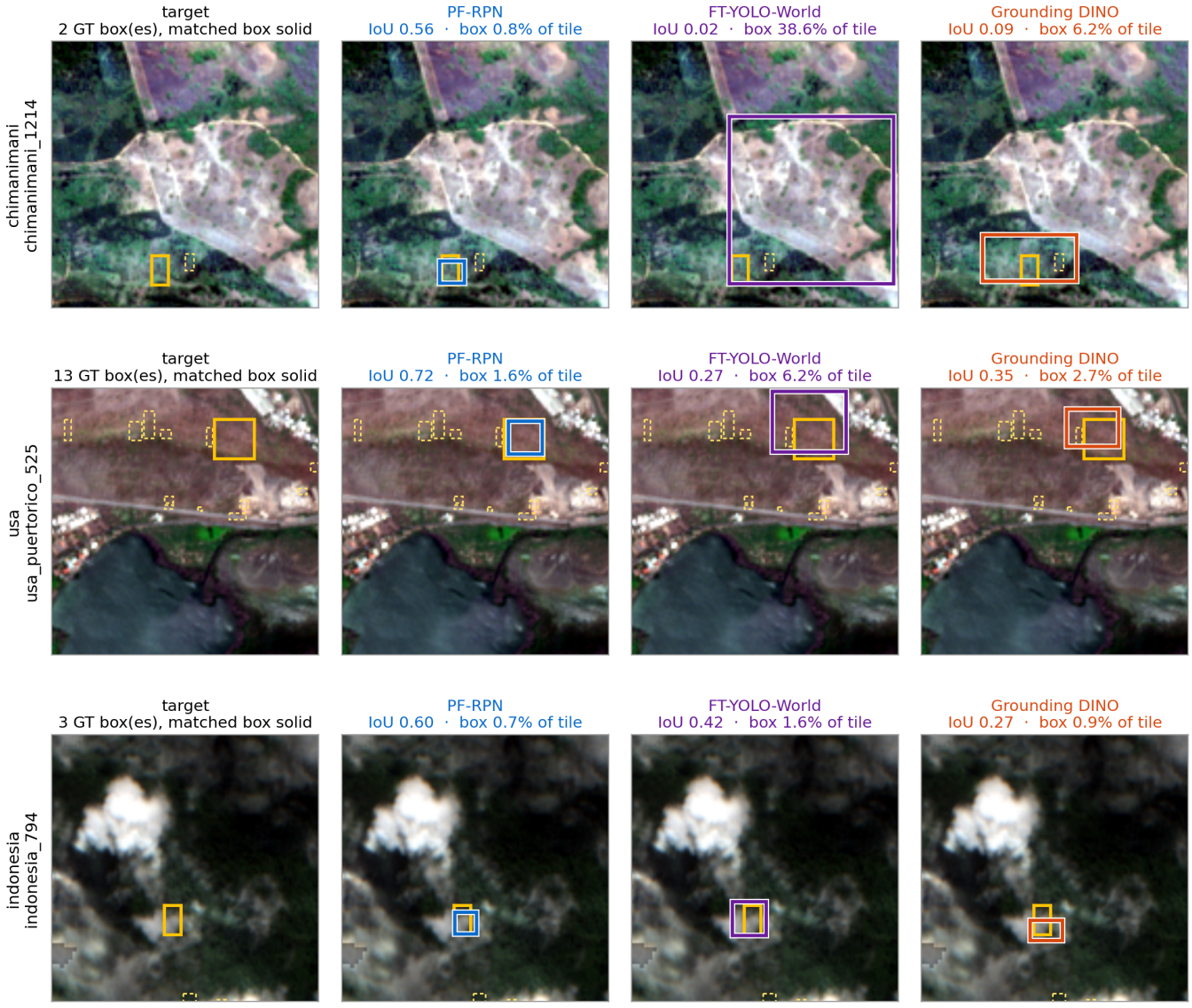


Fig. 9. Three held-out test tiles (one per row; event and tile identifier at left) on which the proposal route satisfies the IoU 0.5 hit criterion and neither detector does. Yellow dashed: ground-truth boxes; solid: the instance used to compute the IoU shown. Blue: best candidate of the proposal route from the shared 900-candidate export; purple: best candidate of fine-tuned YOLO-World, the strongest external arm on this split (0.3321 against 0.1828 for Grounding DINO zero-shot, orange, which is shown for completeness). Segmentation masks are not drawn, because a tile can hold several instances and the stored mask is their union. The proposal boxes cover 0.7-1.6% of the tile, so the recall reported in Section V is not obtained with near-whole-tile boxes; the best detector candidates cover up to 38.6% of the tile and still miss the matched instance.

produces it. The practical consequence is a rule for how the comparison must be run: a weak scorer erases a real paradigm difference, and only a scorer with semantic content, shared between the arms, restores it. What this stage establishes is the comparison itself: with a scorer that carries semantics but none of a detector's own scores, the paradigm difference becomes measurable at a budget a user would actually set, which is the quantity the routing decision needs.

G. Cost and a recommended operating point

The recall ceiling is reached only at a large candidate budget. On the event-level split AR@900 is 0.7059, but the marginal return per additional candidate falls from about 5.5e-3 between 10 and 30 candidates to 1.2e-4 between 500 and 900. At 300 candidates the route already attains 0.5908, 84% of its ceiling, at one third of the budget, so we would recommend an operating point near 300 proposals per image. Inference costs 37.2 ms and 3.8 GB of peak memory per image for the proposal route, against 19.7 ms/1.3 GB for YOLO-World zero-shot and 203.2 ms/2.5 GB for Grounding DINO zero-shot (Fig. 10; see S1-C).

VI. DISCUSSION

Two findings here concern how a comparison is made rather than which detector wins; both are stated in a form that another benchmark can test. First, the choice between a prompt-free proposal route and a trained detector can be anticipated from the data: computed from the change stack and

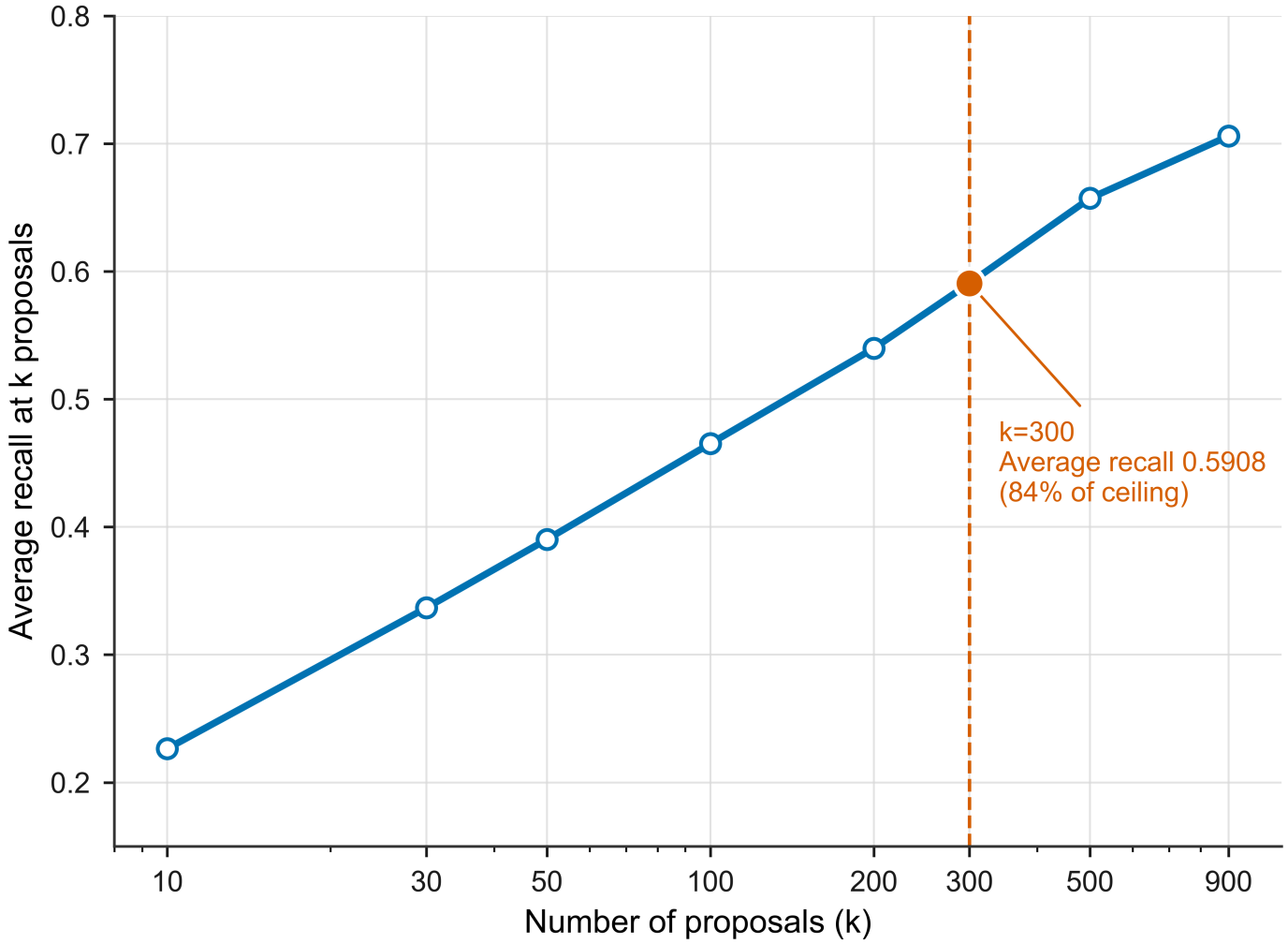


Fig. 10. Candidate budget. Average recall at k proposals for the proposal route on the event-level test split, as a function of the number of proposals per image (logarithmic abscissa); the annotated point marks the recommended operating point at k = 300.

the event's existing masks, the separability statistic separates the ten held-out folds without overlap, leaving the detector with 17.7-26.9% of the route's recall at high separability and 63.7-72.8% at low. It is a decision aid rather than an automatic rule, since it presupposes a reference inventory for the event and its threshold describes these ten folds rather than a calibrated cut. Second, whether a paradigm contrast is visible at a practical budget is a property of the verifier rather than of the paradigms: a scorer without semantic content, or a detector's own confidence, can report a different conclusion from a shared semantic scorer on identical candidates. The subsections below take in turn the contrast both findings rest on (A, B), the change modality the criterion is built from (C), its positioning against transferability measures developed elsewhere (D), the scope of the claims (E), and what follows for practice (F).

A. The comparator sets the size of the contrast, not its direction

Our first result is that recall on held-out events separates the two detector families from a prompt-free proposal route, for a reason that lies in what each family must transfer: the appearance statistics of events already seen, and the class semantics of a name that refers to different surfaces in different regions: the first is what a closed-set landslide detector learns from its training events [13], [15], the second is what a text-conditioned detector grounds in the image [26], [27]. A prompt-free, class-agnostic proposal route, which needs only to localise regions of interest and not to name them, retains substantially more recall on the same imagery and partitions. Because every arm was measured with the same export and scoring procedure, the contrast is between paradigms rather than between implementation details. Its size, however, is not a property of the paradigm alone. On the event-level split, under a matched cap of 900 candidates, the ratio is 2.13-fold against the strongest fine-tuned detector and 3.86-fold against the strongest zero-shot one, and it widens to 7.7-fold at the strict budget of 100 candidates per image, which every arm can fill. On the ten-fold protocol the strongest external arm is Grounding DINO zero-shot, and the proposal route is 2.73-fold above it at 900 candidates and 3.11-fold above fine-tuned YOLO-World. The two-orders-of-magnitude figure obtained against the weak closed-set control was a property of that

export cap rather than of the paradigm. A shared verification stage then reduces the separation at a practical budget to 1.05-fold with statistics-based scorers and restores it to 1.8–2.6-fold once the scorer has semantic content. What survives all three reductions is a consistent ceiling difference: consistently more landslides are present in the prompt-free candidate pool than in the others. This echoes a pattern reported outside remote sensing, where a systematic evaluation found both training-free and supervised detection paradigms brittle under distribution shift [28].

B. Why a prompt-free route survives where grounding does not

A text-conditioned detector must match image content to a description of a landslide [26], [27], and it is the description (not the geometry of the failure) that must transfer across events. When lithology, vegetation or triggering mechanism changes, the same name refers to regions with different appearance and matching degrades. A closed-set detector has the complementary problem: it has learned the appearance statistics of the events it saw and has no mechanism for the statistics of the next one [30]. A class-agnostic proposer avoids both because its objective is to place a region around anything locally object-like and to defer the semantic decision. Within this reading, the advantage we observe comes not from better localisation of landslides specifically but from not requiring a decision that shift makes unreliable. This is a proposal-level argument rather than a uniquely established mechanism. Its central premise, however, is already tested: the proposal route is trained and evaluated with no change information in two of the ablation configurations. With optical, SAR and topography but no temporal channels it reaches AR@900 = 0.4642, against 0.7127 when the change channels are added (three seeds); with optical input alone it still reaches 0.3533 (A0). The advantage over both detector families therefore does not rest on the change modality. What remains untested is the alternative account, that the proposal route is simply more robust to low signal-to-noise input than a detector that must commit to a class; we therefore state the reading as an argument rather than as a measured mechanism. The reading is also consistent with the open-world proposal literature, where removing classification from the proposal stage improves the localisation of objects whose categories were never seen [31].

C. The event-change signal is necessary in every fold; its yield is set by the event

The event-change modality carried most of the accuracy in the event-level ablation, and removing it reduced recall in every cross-setting fold. That the change can be measured at all rests on the per-tile acquisition series this benchmark inherits [37]. Its contribution is nevertheless strongly event-dependent, and only moderately related to separability measured without training. This has a practical consequence for anyone deciding whether to spend acquisition and compute budget on change features: a cheap data-level separability analysis gives a useful but unreliable forecast, because folds with low separability did not always lose little and folds with high separability did not always gain a great deal. That observation qualifies what single-split analyses of change specificity can support [2], and it also complements the finding that additional event-level information improves cross-event performance in global earthquake-triggered landslide prediction [10]: additional temporal information helps, but how much it helps in a given event is not read off a prior diagnostic. The statement our data support is that the change modality is a general source of signal whose yield is event-specific.

The monotone ordering of Table III gives a sharper reading of why the change channels matter. The widened event window is a cleaner supervision signal: it is anchored to the event and 98.5% of its pixels are usable in both frames, against 77.6% for the single adjacent pair. Training on it teaches the change encoder filters that respond to event-driven change. At inference the long baseline supplies a larger contrast, because a scar that appears at the event is still visible 678 days later, and those same filters exploit it better than filters trained on the long baseline itself. Training on the long baseline instead teaches the encoder to also key on seasonal drift, which dilutes the event signal.

This also explains why the post-hoc route fails and why the matched route is not the best one. Applying the event window at inference presents a frozen encoder with content it was not trained to interpret, and evaluating the retrained model on its own training convention removes precisely the additional contrast that made the transfer work.

D. Positioning against recent cross-domain work

GeoPhysAdapter showed that bounded adaptation of a frozen visual backbone can substantially reduce cross-domain error in landslide mapping, and argued that the candidate-body decision unit matters more than the pixel [1]. That reading matches transferability evaluations of segmentation models on this task [3] and the domain-adaptation literature [21], and our result is complementary and, in one respect, more primitive: before an adapter can reduce error, the proposal stage must place the true landslides in the candidate set at all, and we find that this stage is paradigm-dependent enough to separate the detector families by a factor of two to four at a matched 900-candidate budget (7.7-fold at the strictest common budget). The foundation-model adaptation work now appearing on this task [6], [7], [8], [9] operates one level further down the same pipeline: it improves segmentation accuracy on one or a few target areas, and its gains are conditional on a small labelled target set. Our claims do not compete with that line: we ask what a class-agnostic route retains at the discovery stage under event and region isolation, and whether the resulting contrast can be forecast from the data alone. The two findings are consistent — both locate the difficulty in what must transfer rather than in the imagery — and they suggest a division of labour in which a prompt-free proposer supplies candidates and an adapter-driven verifier supplies precision. Our criterion belongs to the family of transferability measures that predict target performance without training on the target [39]–[41]; unlike them it is computed from the event's own mask rather than from source-to-target feature statistics, which is what

makes it usable before any model is fitted. The two statistics are complementary, and testing them in combination is the natural extension of the present test.

Scope of the claims Several boundaries qualify these conclusions. First, the protocols isolate the event and region dimensions: an entire region is absent from training in the LORO folds, and the LORO means are not lower than the LOEO means. What they do not isolate is the source (the benchmark spans one sensor pair and one annotation protocol, both inherited from the released dataset [37], [38]) so the paradigm contrast is established across events and regions within this benchmark, which is the scope of the claim throughout. Second, the cross-setting folds were run with a single seed whereas the event-level comparison used three, and the pipeline is not numerically deterministic: rerunning one fold reproduces the archived value to 0.0013 AR, and adding two seeds to each of three folds gives fold-wise run-to-run deviations of 0.0076, 0.0234 and 0.0251 AR against a between-fold standard deviation of 0.159. Run-to-run variability therefore stays more than six times below the between-fold dispersion, and the two main effects exceed the largest observed deviation by factors of 14 and 27-29, but the per-fold change-modality contrasts (0.031-0.30 AR) are not all resolvable on a single seed; the smallest is reported as a paired same-seed comparison (Section V-B and Supplementary S1). Third, the verification stage is deliberately shared and paradigm-neutral, so it measures how far the recall ceiling can be recovered rather than the performance of a deployable inventory: its absolute level is low (about 3% of landslides recovered within ten proposals per image), and precision at a usable operating point is not demonstrated. Fourth, proposal scores occupy a narrow range, which is why the reported measure is recall at a fixed proposal budget; conclusions about precision at low false-positive rates are not supported.

A further boundary concerns the labels themselves. Of 4,392 positive tiles, 28.5% are multi-event tiles for which the dating module returns not one window but a set of confidence-ranked candidate dates, between 2 and 11. Their median date confidence is 0.44 to 0.61 against 1.00 for single-event tiles, and their median ground-truth pixel count rises monotonically with the number of candidates, from 245 pixels at two candidates to 2,510 at eleven. The tiles with the least certain dates are therefore the ones containing the largest landslides, and the size-stratified results show the highest recall in the large bucket. Part of that advantage reflects insensitivity to the window rather than detection capability alone. For multi-event tiles we use the highest-confidence candidate, which is the only deterministic rule the dataset supports. (see S1-G).

A second protocol does not reproduce the gain. When the whole Indonesian region is held out and the model is retrained on the other ten inventories, the same intervention lowers AR@900 from 0.7891 to 0.7717 and AR@300 from 0.7154 to 0.6731 (paired over the 253 annotated images: -0.0173, 95% CI [-0.041, +0.006], and -0.0423, [-0.067, -0.018]); at k=100 the loss is 0.0535 with an interval that excludes zero. The comparison is tight because the event-level split scores the same 253 tiles: there the same intervention gains 0.0414 at AR@900 and 0.0426 at AR@100, so what separates the two protocols is the backing model rather than the region, whose calendar and windows are identical in both (see S1-F4). The gain is therefore stated as a result obtained under the event-level protocol with its own anchor, and its size is reported as protocol-specific. What carries across protocols is the interval itself: it is recoverable from the calendar and it is unrecoverable after training, and both statements hold on the region-level fold as well. The size of the gain is the part that must be verified on the protocol a user intends to deploy, and we say so explicitly rather than leave it to be inferred. The headroom reading is the one we tested and pre-registered, and rejected: on the fold with the weakest archived anchor (Italy, AR@900 0.2896, re-scored at 0.2950 on this machine) the same intervention loses 0.0648, 0.0584 and 0.0779 at AR@900 over seeds 42, 123 and 777, with every paired interval below zero, so the gain is not a simple function of the room a baseline leaves (see S1-J and S1-K).

Two further limits apply to the interval result specifically. It is established on one dataset and one split family; the interval gain in Table III is averaged over three seeds, and a single cross-protocol fold is reported in the supplementary material. And the training arm uses a fixed budget of eight epochs, so the gain is measured at that budget rather than at convergence. (see S1-I).

E. What the result enables

Taken together, the findings support a narrower but firmer recommendation than the effect sizes first suggest. The first stage of automated landslide mapping should be designed for localisation rather than recognition [31]: a prompt-free proposal front end places substantially more landslides into a fixed candidate budget than either family of detector, and the difference survives every correction we are able to apply. What that candidate set is then worth depends on how it is reduced, and that step is not yet solved: with the verifiers we tested, ten proposals per image recover 1.9–2.9% of landslides, and the separation between paradigms at that budget ranges from 1.05-fold with a statistics-based scorer to 1.8–2.6-fold with a scorer that has semantic content. Acquisition planning, by contrast, can be guided before any model is fitted, because the separability of the change signal forecasts which paradigm will win on a held-out event. The remaining uncertainty therefore lies in the verification stage rather than in the proposal stage.

F. Future work

Five open items follow directly from this study. The first is external validity: the two-regime separation is established on ten held-out folds of one benchmark, so the immediate test is whether it reappears on an independently annotated corpus, and whether a threshold fitted on one corpus transfers to another; until then the criterion is stated as a benchmark-specific result rather than a general one. The second is prospective use: fixing the statistic on a training corpus and applying it to events whose outcome is not yet known would convert a retrospective fit into a decision rule. The third is the verifier: our verification stage shows that a scorer without semantic content removes a real paradigm difference, and what does

not yet exist is an arm-agnostic semantic scorer that can be reused across benchmarks instead of being retrained per study. The fourth is breadth: the comparison should be extended to proposal front ends other than the one adopted here [4], [30], because the criterion forecasts a contrast between paradigms rather than the behaviour of one implementation. Until that is done, every statement about the proposal paradigm in this paper should be read as a statement about PF-RPN as its representative, the role it is explicitly assigned in Section IV-A. The fifth is reporting practice: per-fold contrasts of the same size as run-to-run variability need paired same-seed replicates, the design we apply here to the smallest effect, and we would expect future comparisons at the proposal stage to report per-arm candidate budgets and both image-averaged and ground-truth-weighted recall as a matter of course, in the spirit of the protocol papers that standardise a whole task family [22]. (see S1-E4).

VII. CONCLUSION

Candidate discovery from satellite imagery is decided by two choices that current practice leaves implicit: which route takes it, and which two acquisitions are differenced. We compared three paradigms at equal candidate budgets under event- and region-isolated protocols, and gave a training-free criterion that predicts the winning route from a cheap property of the imagery, with the correct sign in ten of ten folds. We then separated two decisions that are usually conflated. The event interval is a calendar fact, recovered exactly in 97.9% of 3,135 tiles from 11 inventories without annotation, and it is committed during training rather than at inference. Committing it during training, while leaving the deployed long baseline unchanged, raises AR@900 from 0.7094 to 0.7474 over three seeds and strict-budget recall by 11.7% relative, at no inference cost and without architectural change.

The wider lesson is that the interval belongs to the specification of the task rather than to the tuning of the model, and that the training and deployment intervals are separate choices. Reporting them explicitly, together with the candidate budget of every arm, would make comparisons in this literature cheaper and directly comparable.

SUPPLEMENTARY MATERIAL

Sections S1-A to S1-K and Tables S1-B to S1-I, cited throughout this paper, are provided in the accompanying supplementary document. The appendix below summarises the run-to-run noise floor that several of those sections rely on.

Reproducibility and the run-to-run noise floor

We separate three reproducibility levels.

(i) Data. The dataset was rebuilt independently from the public source and checked at three levels: a randomly selected positive tile (chimanimani_1004) is SHA256-identical to the archived copy in all six of its .npy arrays; the train, validation, test, seven leave-one-event-out and three leave-one-region-out manifests reference no missing tiles; and the training-set normalisation statistics recompute exactly (6,813 tiles, 111,624,192 pixels, with all 15 channel means and standard deviations bit-identical to the archived values). The data layer therefore contributes no difference between runs.

(ii) Training. The training pipeline is not forced to be deterministic: non-deterministic GPU kernels (atomic accumulation and reduction order under mixed precision) produce small differences in weights between runs. Re-running the italy leave-one-event-out fold end-to-end under the published protocol gave AR@900 = 0.2906 against the archived 0.2893 (difference +0.0013), AR@100 = 0.1408 against 0.1425 (difference -0.0017) and AR@300 = 0.2005 against 0.2029 (difference -0.0024), on an identical evaluation set (4,236 images, 35,293 ground-truth boxes). The envelope is therefore fold-wise rather than constant: adding two seeds to each of three folds gives maximum deviations from the same-configuration re-run of 0.0076 AR (italy, the lowest fold), 0.0234 AR (chimanimani) and 0.0251 AR (indonesia, the highest fold), with a largest run-to-run standard deviation of 0.0104 against a between-fold standard deviation of 0.159.

(iii) Conclusions. The reported effects separate into two classes by size. The two main effects exceed the largest observed run-to-run deviation (0.0251 AR) by more than an order of magnitude: the modality ablation spans AR@900 = 0.3533 to 0.7167 (2.03-fold; 0.36 AR, 14 times), and under cross-event (LOEO) and cross-region (LORO) evaluation the closed-set control, exported at fewer than 100 candidates per image, reached 0.0030 and 0.0027 while the prompt-free proposal route reached 0.675 and 0.726, gaps of 0.67 and 0.72 AR (27 and 29 times). Single-fold contrasts of the size of the change-modality contribution (0.031-0.30 AR) are not all of that size relative to run-to-run noise; those that are not are reported with their seed structure, and the smallest was therefore adjudicated by a paired same-seed re-run: across three seeds the change-modality gain is 0.024 ± 0.006 AR, with all three paired differences positive. No conclusion in this work rests on a difference smaller than the run-to-run variation of the comparison it is drawn from.

REFERENCES

[1] Z. Liu, M.-P. Kwan, J. Wu, and H. Li, "GeoPhysAdapter: Scale-matched geophysical adaptation for cross-domain landslide mapping with vision foundation models," *ISPRS J. Photogramm. Remote Sens.*, vol. 243, pp. 140–159, 2026, doi: 10.1016/j.isprsjprs.2026.09.035.

[2] O. Ghorbanzadeh et al., "The outcome of the 2022 Landslide4Sense competition: Advanced landslide detection from multisource satellite imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9927–9942, 2022, doi: 10.1109/JSTARS.2022.3220845.

[3] O. Ghorbanzadeh, A. Crivellari, P. Ghamisi, H. Shahabi, and T. Blaschke, "A comprehensive transferability evaluation of U-Net and ResU-Net for landslide detection from Sentinel-2 data," *Sci. Rep.*, vol. 11, no. 1, art. no. 14629, 2021, doi: 10.1038/s41598-021-94190-9.

[4] Q. Tang, C. Liu, S. Zhang, et al., "Prompt-free universal region proposal network," arXiv:2603.17554, 2026.

[5] Z. Li, X. Gao, Z.-Y. Chen, Y. Duan, and F. Yang, "Prompt-free efficient adaptation of segment anything model for remote sensing landslide detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 7, pp. 10564–10576, 2026, doi: 10.1109/TCSVT.2026.3686175.

[6] L. Zhang, X. Liu, J. Zhang, et al., "MTGF-SAM: Multilevel terrain-gated fusion of Segment Anything Model for landslide detection in remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 33258–33275, 2026, doi: 10.1109/JSTARS.2026.3734278.

[7] Z. Fu, F. Wang, Z. Tu, et al., "LSDSAM: Harnessing visual foundation model and enhanced transfer learning toward practical landslide detection in few-shot scenarios," *IEEE J. Sel. Topics Appl. Earth*

Observ. Remote Sens., vol. 19, pp. 12344–12365, 2026, doi: 10.1109/JSTARS.2026.3678141.

[8] C. Hou, J. Yu, D. Ge, et al., "A transfer learning approach for landslide semantic segmentation based on visual foundation model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 11561–11572, 2025, doi: 10.1109/JSTARS.2025.3559884.

[9] Y. Fan, H. Zhang, J. Chen, et al., "Cross-regional few-shot landslide segmentation with visual foundation model via frequency decoupling and iterative refinement," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 21646–21660, 2026, doi: 10.1109/JSTARS.2026.3706738.

[10] X. Fan et al., "Deep learning can predict global earthquake-triggered landslides," *Natl. Sci. Rev.*, vol. 12, no. 7, art. no. nwaf179, 2025, doi: 10.1093/nsr/nwaf179.

[11] S. L. Ullo, A. Mohan, A. Sebastianelli, et al., "A new Mask R-CNN-based method for improved landslide detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3799–3810, 2021, doi: 10.1109/JSTARS.2021.3064981.

[12] H. Cai, T. Chen, R. Niu, et al., "Landslide detection using densely connected convolutional networks and environmental conditions," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 5235–5247, 2021, doi: 10.1109/JSTARS.2021.3079196.

[13] T. Liu, T. Chen, R. Niu, et al., "Landslide detection mapping employing CNN, ResNet, and DenseNet in the Three Gorges Reservoir, China," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11417–11428, 2021, doi: 10.1109/JSTARS.2021.3117975.

[14] T. Chen, X. Z. Gao, G. Liu, et al., "BisDeNet: A new lightweight deep learning-based framework for efficient landslide detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3648–3663, 2024, doi: 10.1109/JSTARS.2024.3351873.

[15] Y. Yang, Z. Miao, Z. Hua, et al., "Lightweight attention-guided YOLO with level set layer for landslide detection from optical satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3543–3559, 2024, doi: 10.1109/JSTARS.2024.3351277.

[16] Y. Chen, D. Ming, J. Yu, et al., "Susceptibility-guided landslide detection using fully convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 998–1018, 2022, doi: 10.1109/JSTARS.2022.3233043.

[17] P. Lv, L. Ma, Q. Li, et al., "ShapeFormer: A shape-enhanced vision transformer model for optical remote sensing image landslide detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 2681–2689, 2023, doi: 10.1109/JSTARS.2023.3253769.

[18] J. Cai, M. Ding, M. Ullah, et al., "Learning discriminative representations of landslides via deep metric learning under limited supervision," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 31550–31569, 2026, doi: 10.1109/JSTARS.2026.3731892.

[19] Y. Yi and W. Zhang, "A new deep-learning-based approach for earthquake-triggered landslide detection from single-temporal RapidEye satellite imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 6166–6176, 2020, doi: 10.1109/JSTARS.2020.3028855.

[20] J. Luo, T. Luo, M. Wang, et al., "Learning domain generalized remote sensing image segmentation by multiscale instance disentanglement," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 13538–13553, 2025, doi: 10.1109/JSTARS.2025.3568853.

[21] J. Peng, Y. Huang, W. Sun, et al., "Domain adaptation in remote sensing image classification: A survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9842–9859, 2022, doi: 10.1109/JSTARS.2022.3220875.

[22] G. Vivone, M. Dalla Mura, A. Garzelli, et al., "A benchmarking protocol for pansharpening: Dataset, preprocessing, and quality assessment," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6102–6118, 2021, doi: 10.1109/JSTARS.2021.3086877.

[23] Q. Zhu, Y. Fang, Y. Cai, et al., "Rethinking scanning strategies with vision Mamba in semantic segmentation of remote sensing imagery: An experimental study," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 18223–18234, 2024, doi: 10.1109/JSTARS.2024.3472296.

[24] L. Jiang, F. Wang, W. Zhang, et al., "Rethinking the key factors for the generalization of remote sensing stereo matching networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 4936–4948, 2024, doi: 10.1109/JSTARS.2024.3524078.

[25] G. Yao, W. Zhou, M. Liu, et al., "An empirical study of the convolution neural networks based detection on object with ambiguous boundary in remote sensing imagery—A case of potential loess landslide," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 323–338, 2021, doi: 10.1109/JSTARS.2021.3132416.

[26] S. Liu, Z. Zeng, T. Ren, et al., "Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024, pp. 38–55, doi: 10.1007/978-3-031-72970-6_3.

[27] T. Cheng, L. Song, Y. Ge, et al., "YOLO-World: Real-time open-vocabulary object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 16901–16911, doi: 10.1109/CVPR52733.2024.01599.

[28] J. Sandhan, H. Jaiswal, F. Cheng, and Y. Murawaki, "Can we trust LLM detectors?," in *Proc. ANLP*, 2026, arXiv:2601.15301.

[29] C. Li, Q. Zhang, X. Liang, Y. Xiao, S. Li, and P. Wang, "BAPA-Net: Cross-domain few-shot object detection in remote sensing via background-aligned prototype aggregation," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, art. no. 5514618, 2026, doi: 10.1109/TGRS.2026.3696309.

[30] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, 2017, doi: 10.1109/TPAMI.2016.2577031.

[31] D. Kim, T.-Y. Lin, A. Angelova, I. S. Kweon, and W. Kuo, "Learning open-world object proposals without learning to classify," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 5453–5460, 2022, doi: 10.1109/LRA.2022.3146922.

[32] S. Tian, Z. Huang, L. Cao, et al., "Pi-noise-guided student–teacher distillation detector for open-vocabulary aerial object detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 30368–30389, 2026, doi: 10.1109/JSTARS.2026.3727402.

[33] W. Xu, C. Wang, X. Feng, et al., "Generalization boosted adapter for open-vocabulary segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 1, pp. 520–533, 2024, doi: 10.1109/TCSVT.2024.3454227.

[34] D. Wang, J. Zhang, M. Xu, et al., "MTP: Advancing remote sensing foundation model via multitask pretraining," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11632–11654, 2024, doi: 10.1109/JSTARS.2024.3408154.

[35] Z. Gao, S. Sun, M. Cheng, et al., "Could describe anything model understand remote sensing objects?," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 26586–26598, 2025, doi: 10.1109/JSTARS.2025.3616330.

[36] H. Gao, S. Wu, Y. Wang, et al., "FSOD4RSI: Few-shot object detection for remote sensing images via features aggregation and scale attention," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 4784–4796, 2024, doi: 10.1109/JSTARS.2024.3362748.

[37] P. Hoehn, K. Heidler, R. Behling, and X. X. Zhu, "A spatio-temporal dataset for satellite-based landslide detection," *Sci. Data*, vol. 12, no. 1, art. no. 1772, 2025, doi: 10.1038/s41597-025-06167-2.

[38] Y. Xu, C. Ouyang, Q. Xu, et al., "CAS landslide dataset: A large-scale and multisensor dataset for deep learning-based landslide detection," *Sci. Data*, vol. 11, no. 1, art. no. 12, 2024, doi: 10.1038/s41597-023-02847-z.

[39] C. V. Nguyen, T. Hassner, M. Seeger, and C. Archambeau, "LEEP: A new measure to evaluate transferability of learned representations," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2020, pp. 7294–7305.

[40] K. You, Y. Liu, M. Long, and J. Wang, "LogME: Practical assessment of pre-trained models for transfer learning," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 12133–12143.

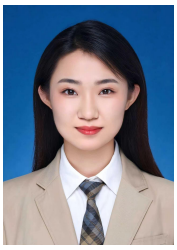
[41] M. Pándy, A. Agostinelli, J. Uijlings, V. Ferrari, and T. Mensink, "Transferability estimation using Bhattacharyya class separability," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 9162–9172, doi: 10.1109/CVPR52688.2022.00896.



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